Artificial Intelligence Review (2026) 59:203
https://doi.org/10.1007/s10462-026-11608-4

Deepfake detection across image, video, and audio: a comprehensive survey with empirical evaluation of generalization and robustness

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Received: 30 September 2025 / Accepted: 28 May 2026
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Abstract
Deepfakes (DFs) have emerged as a significant threat in recent years. They are exploited for malicious purposes such as impersonation, misinformation dissemination, and artistic style imitation, raising critical ethical and security concerns. This survey presents a comprehensive cross-modality analysis of passive DF detection, examining image, video, and audio modalities simultaneously. Distinct from multimodal detection which fuses multiple data streams (e.g., audio-visual, text-visual), our cross-modality analysis investigates the interconnected relationships, shared methodological principles, and common vulnerabilities across independent modalities. We systematically categorize detection approaches based on their underlying methodologies: forensic-based, data-driven, fingerprint-based, and hybrid techniques for visual modalities, and handcrafted versus learnable features for audio. We also extend our analysis beyond mere detection accuracy to include essential performance dimensions for real-world deployment, including generalization and robustness. Additionally, this survey provides a unified evaluation protocol across 10 popular datasets to assess detection accuracy, generalization, and robustness in each modality. Specifically, we conduct extensive empirical evaluations across three critical dimensions: (1) verification of reported within-domain accuracy for 13 unimodal detectors, (2) cross-domain generalization assessment of 33 methods specifically designed to enhance generalization capability, and (3) robustness evaluation of 6 methods against adversarial attacks. Our experiments reveal a persistent generalization gap, with performance degradations of $10 - 15\%$ in out-of-distribution scenarios, and vulnerability to white-box adversarial attacks exceeding 80% success rates. We also analyze the advantages and limitations of existing datasets, benchmarks, and evaluation metrics for passive DF detection. Finally, we propose future research directions to address these unexplored and emerging issues in the field of passive DF detection. This survey serves as a comprehensive resource for researchers and practitioners, providing insights into the current landscape, methodological approaches, and promising future directions in this rapidly evolving field.

Keywords Deepfake detection · Generalization · Robustness · Empirical evaluation · Cross-modality analysis

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Published online: 12 June 2026

1 Introduction

The term *DeepFake* (DF) describes synthetic media (e.g., image, video, audio) produced by *Generative Models* (GMs) for malicious purposes, causing serious threats across financial, political, and social domains. In February 2024, a sophisticated DF attack on Arup Group caused a $25.6 million loss when the attacker used AI-generated avatars of the CEO in a video conference (Chen and Magramo 2024). Similarly, in Singapore, a coordinated campaign impersonating political leaders defrauded 12,000 investors of $180 million through synthetic endorsements of cryptocurrency schemes (Natnicha 2024). Beyond financial fraud, DF have been used for electoral interference where 23,000 voters received AI-generated robocalls that mimic President Biden's voice patterns to suppress voter turnout (Guardian 2024).

In response to these emerging threats, two main categories of DF detection approaches have been developed: proactive and passive. Proactive approaches counteract DF manipulation before it occurs. Techniques include adding perturbations to real data or embedding watermarks during generation (Nguyen-Le et al. 2025). In contrast, passive approaches are popular and straightforward solutions against DFs as they are cast as binary classification problem. Passive approaches detect DFs after they are generated, focusing on analyzing intrinsic properties of them to distinguish between authentic and synthetic content. In this work, we focus on passive approaches rather than proactive ones.

1.1 Motivation

This survey is motivated by three fundamental imperatives that are essential for advancing passive DF detection research toward practical and deployable solutions.

First, understanding inter-modality relationships is crucial for developing unified detection strategies. The current landscape of DF detection research suffers from a fragmented approach where image, video, and audio detection methods are developed and evaluated independently. This isolation fails to recognize that DF artifacts manifest differently across modalities, yet there may be underlying commonalities that can be exploited. Image-based detectors achieving near-perfect accuracy are often ineffective when applied frame-by-frame to videos. This is because they fail to capture critical temporal inconsistencies, including unnatural head movements, flickering artifacts, and inter-frame coherence violations (Liu et al. 2023; Gu et al. 2021; Yin et al. 2023). Similarly, audio DF exhibit entirely distinct artifact patterns, such as spectral anomalies, phase coherence irregularities, and prosodic inconsistencies. These patterns require specialized detection architectures (Sun et al. 2023; Yan et al. 2022). By exploring these inter-modality relationships, we can identify shared patterns that transcend individual modalities, and pave the way for more robust and versatile detection frameworks.

Second, existing surveys fundamentally fail to address critical properties required for real-world deployment beyond accuracy metrics. Current survey papers (Mirsky and Lee 2021; Malik et al. 2022; Wang et al. 2022a, b; Pei et al. 2024; Kaur et al. 2024; Yi et al. 2023) primarily focus on reviewing detection methods based on their accuracy performance within the same dataset (within-domain evaluation), which achieves high accuracy (often exceeding 95%). However, these evaluations do not account for the diverse challenges encountered in real-world scenarios:

- *Generalization capability* is a critical concern as detectors trained on specific datasets often fail to maintain performance when exposed to new data distributions or new manipulation techniques (Corvi et al. 2023; Ojha et al. 2023; Corvi et al. 2023; Chen et al. 2022). Existing surveys lack the evaluation of detectors' performance on cross-domain scenarios (Kaur et al. 2024; Malik et al. 2022; Wang et al. 2022a; Pei et al. 2024). This gap highlights the need for a comprehensive survey and evaluation of existing generalization methods.
- *Robustness to adversarial and distortion attacks* represents another critical dimension. Existing surveys do not examine how detection methods withstand adversarial attacks designed to evade detection (Corvi et al. 2023; Hou et al. 2023). They also neglect post-processing operations that degrade detection performance (Feng and Owens 2023; Xu et al. 2024; Le and Woo 2023). This oversight leaves researchers and practitioners without guidance on the resilience of current methods against real-world manipulations.*Third, existing surveys on passive DF detection predominantly focus on analyzing methods conceptually, with limited empirical evaluation*. Current survey papers (Kaur et al. 2024; Malik et al. 2022; Wang et al. 2022a; Pei et al. 2024) provide extensive taxonomies and theoretical discussions. However, they lack empirical evaluations that would enable researchers to understand the relative strengths and limitations of different approaches. Without comparative experiments, it is impossible to assess whether reported improvements represent genuine advances or merely result from favorable experimental setups. This survey addresses this critical gap by providing comprehensive empirical evaluation of current methods across three essential protocols: (1) in-distribution (within-domain) evaluation to establish baseline performance, (2) out-of-distribution (cross-domain) evaluation to assess generalization capabilities, and (3) robustness evaluation against adversarial attacks. These motivations underscore the need for this comprehensive survey. Therefore, in this work, we aim to provide a unified cross-modality analysis of detection methods across modalities, review approaches that address generalization and robustness challenges, and conduct empirical experiments to validate and compare existing approaches.

1.2 Related survey work

Previous surveys have typically reviewed single modality in isolation, including image (Mirsky and Lee 2021; Malik et al. 2022; Wang et al. 2022a; Pei et al. 2024; Wang et al. 2022b), video (Malik et al. 2022; Pei et al. 2024; Kaur et al. 2024), and audio (Yi et al. 2023; Li et al. 2025). Table 1 compares our work with existing survey papers, highlighting the strengths and drawbacks of each work.

Mirsky and Lee (2021) (Mirsky and Lee 2021) provides comprehensive insights into GAN architectures employed in DF generation and Pei et al. (2024) (Pei et al. 2024) focused more on benchmarks for evaluating GMs. Wang et al. (2022a) (Wang et al. 2022a) reviews detection techniques specifically for identifying GAN-generated artifacts, encompassing DL-based, physical-based, and physiological-based methods, while Malik et al. (2022) (Malik et al. 2022) provides broader categories of DF detection approaches in both image and video modalities. Xie et al. (2025) (Xie et al. 2025) provides a comprehensive review of visual DF detection methods in the era of AI-generated content (AIGC). This work highlights that existing detectors' performance drop significantly when evaluated on datasets

Table 1 Comparison of this work with existing DF detection surveys

Surveys	Year	Modality	Generation Techniques	Detection Methods	Inter-modality Analysis	Empirical Evaluation			Dataset & Benchmark Analysis
						Detection accuracy	Generalization	Robustness	
The creation and detection of deepfakes: A survey (Mirsky and Lee 2021)	2021	Image	✓✓	✓	×	×	×	×	×
Deepfake detection for human face images and videos: A survey (Malik et al. 2022)	2022	Image and Video	✓✓	✓	×	×	×	×	×
Gan-generated faces detection: A survey and new perspectives (Wang et al. 2022a)	2022	Image	✓	✓✓	×	×	×	×	×
Audio deepfake detection: A survey (Yi et al. 2023)	2023	Audio	✓	✓✓	×	×	×	×	×
A survey on speech deepfake detection (Li et al. 2025)	2025	Audio	✓	✓✓	×	✓	✓	×	×
Deepfakes generation and detection: state-of-the-art, open challenges, countermeasures, and way forward (Masood et al. 2023)	2024	Unimodal	✓✓	✓✓	×	×	×	×	×
A survey on the detection and impacts of deepfakes in visual, audio, and textual formats (Mubarak et al. 2023)	2024	Unimodal	✓	✓	×	×	×	×	×
Deepfake generation and detection: a benchmark and survey (Pei et al. 2024)	2024	Image and Video	✓	✓	×	×	×	×	∼
Deepfake video detection: challenges and opportunities (Kaur et al. 2024)	2024	Video	×	✓✓	×	×	×	×	×
Evolving from single-modal to multimodal facial deepfake detection: a survey (Liu et al. 2024)	2024	Unimodal and multimodal	✓	✓✓	×	×	×	×	✓
Deepfake detection: A comprehensive study from the reliability perspective (Wang et al. 2022b)	2024	Image	×	✓✓	×	×	×	✓✓	∼
Fake news detection landscape: datasets, data modalities, AI approaches, their challenges, and future perspectives (Hussain et al. 2025)	2025	Unimodal and Multimodal	×	✓	×	×	×	×	✓✓
DeepFake detection in the AIGC era: A survey, benchmarks, and future perspectives (Xie et al. 2025)	2025	Image and Video	✓	✓✓	✓✓	✓	✓✓	×	✓
A systematic literature review on deepfake detection techniques (Sharma et al. 2025)	2025	Unimodal	✓	✓✓	×	✓	×	×	✓
Ours	2025	Unimodal	×	✓✓	✓✓	✓✓	✓✓	✓✓	✓✓

Notes:✓✓: Comprehensive coverage; ✓: Partial coverage; ∼: Limited discussion; ×: Not covered

generated by DMs. (Yi et al. 2023) is the first work that reviews existing approaches for detecting fake audios, while Li et al. (2025) (Li et al. 2025) extends this by reviewing and evaluating optimization techniques applied in the model training process (e.g., data augmentation), activation functions, and loss functions. Instead of reviewing state-of-the-art (SoTA) detection approaches, the work (Kaur et al. 2024) examines real-world applications of DF video detectors, particularly focusing on computational complexity and scalability considerations. Additionally, Wang et al. (2022b) (Wang et al. 2022b) contributes valuable insights into the reliability aspects of image-based DF detection methods. Recent surveys have reviewed single-modal DF detection approaches (Masood et al. 2023; Mubarak et al. 2023; Sharma et al. 2025) or both single- and multi-model approaches (Liu et al. 2024; Hussain et al. 2025).

As shown in Table 1, existing surveys primarily focus on reviewing generation techniques and detection methods, with limited attention to critical aspects such as generalization, robustness, inter-modality analysis, empirical evaluation, and dataset analysis. Beyond the work (Wang et al. 2022b)'s full coverage of robustness, no existing survey provides systematic categories of generalization and robustness capabilities of DF detectors. Furthermore, prior works rely exclusively on reported results from original papers without independent evaluation under various conditions.

Our survey aims to fill these gaps by providing a comprehensive cross-modality review of passive DF detection methods, with a strong emphasis on generalization and robustness. We also conduct extensive empirical evaluations to validate and compare existing approaches (52 approaches across modalities).

1.3 Contributions

In summary, the main contributions of our work are as follows:

- *Comprehensive cross-modality review with real-world focus*: We systematically examine passive DF detection across image, video, and audio modalities. We analyze inter-modality relationships and critical real-world requirements, including generalization to novel forgery methods and robustness against adversarial and distortion attacks.
- *Extensive empirical evaluation*: We conduct experiments evaluating SoTA passive DF detection methods across three protocols:
  - *Reported performance verification*: We independently verify the accuracy of 13 representative unimodal detectors (8 visual and 5 audio methods) to validate baseline detection capabilities reported in original papers.
  - *Generalization evaluation*: We evaluate 33 methods specifically designed to enhance cross-domain generalization.
  - *Robustness evaluation*: We assess 6 methods that claim to improve robustness against adversarial attacks under white-box, black-box, and transferable attack settings.
- *Dataset limitation analysis*: We identify critical gaps in current benchmark datasets, including class imbalance, and insufficiently challenging scenarios (multiple faces, occlusions).

- *Emerging challenges and future directions*: We synthesize key unexplored areas including detection under challenging conditions, computational efficiency for real-time deployment, privacy-preserving mechanisms, and adaptation to rapidly evolving GMs.

1.4 Survey scope

It is essential to distinguish between multimodal detection and the cross-modality analysis presented in this work. In the field of DF detection, *multimodal* refers to detection approaches that aim to analyze two modalities simultaneously to distinguish between real or fake, such as visual-audio fusion or text-visual analysis (Liu et al. 2024; Hussain et al. 2025). In contrast, this survey conducts a *cross-modality analysis*, which we define as the comparative study of independent modalities (image, video, and audio) under a unified analytical framework. By reviewing and evaluating these modalities side-by-side, we aim to uncover shared challenges, such as the trade-off between generalization and robustness that transcend individual modalities (See Section 6).

Furthermore, although this survey conceptually examines the implications of emerging generation techniques such as Diffusion Models (DMs) and complex real-world conditions, including occlusions and multi-face forgeries, our empirical evaluation deliberately focuses on conventional benchmark datasets. This design choice allows us to establish a reproducible and comparable baseline for existing detectors under a unified protocol. Since many publicly available methods were originally developed for GAN-era or conventional manipulation pipelines, evaluating all of them directly on diffusion-based datasets without comprehensive retraining would impose an extreme zero-shot setting and may not fairly reflect their intended capabilities. Therefore, the empirical results should be interpreted as a standardized conventional benchmark, while diffusion-based and realistic-scenario evaluation remains an important direction for future benchmark studies.

1.5 Survey methodology

This survey follows a systematic approach to select relevant literature on passive DF detection and ensure reproducibility of our literature review process.

1.5.1 Literature search strategy

Databases. We utilize multiple academic databases to ensure comprehensive coverage of relevant literature, including IEEE Xplore, ACM Digital Library, Google Scholar, and arXiv. We only select papers published in peer-reviewed journals or conferences regarding AI and security fields, such as CVPR, ICCV, ECCV, NeurIPS, ICML, AAAI, IJCAI, ICASSP, Interspeech, ACM CCS, IEEE TIFS, IEEE T-PAMI, IEEE T-NNLS.

Search strings. We employed a systematic Boolean search strategy combining primary keywords with modality-specific and methodology-specific terms. Table 2 summarizes our search taxonomy and keywords, organized into four categories: (1) Core detection terms, (2) Modality-specific terms, (3) Methodology-specific terms, (4) Real-world properties.

Our complete search strings were constructed using the following formula:

(Core Detection Terms) AND (Modality Terms) AND (Methodology/Property Terms)

Table 2 Search taxonomy and keywords

Category	Primary keywords	Synonyms and related terms
Core detection terms	Deepfake detection	Synthetic media detection, AI-generated content detection, forgery detection, media forensics
	Deepfake	Synthetic media, AI-generated media, generative media, fake content
Modality-specific terms	Image deepfake	GAN-generated image, diffusion-generated image, fake image
	Face forgery	Face manipulation, face synthesis, synthetic faces, face swap, face reenactment
	Video deepfake	Synthetic video, manipulated video
	Audio deepfake	Synthetic speech, voice synthesis, spoofed speech
	Voice cloning	Voice conversion, text-to-speech
Methodology-specific terms	Supervised learning	Self-supervised learning, unsupervised learning, contrastive learning, metric learning, meta learning, representation learning
	Deep learning	Neural networks
	Forensic-based	Physiological signals, biological signals
	Data-driven	End-to-end learning, feature learning, classification, segmentation, reconstruction
	Fingerprint-based	Generative fingerprints, upsampling artifacts, vocoder fingerprints, architectural artifacts
	Hybrid	Spatial and frequency fusion, spatial and noise fusion
Real-world properties	Generalization	Cross-domain, domain shift, out-of-distribution, unseen data
	Robustness	Adevrsarial examples, adversarial attacks, adversarial traning, distortion attacks
	Post-processing techniques	Compression,degradation, blur, rotation, quality reduction

For example, to search for image DF detection methods that utilize deep learning and address generalization, we used the following search string: ("deepfake detection" OR "synthetic image detection") AND ("image deepfake" OR "GAN-generated image" OR "diffusion-generated image") AND ("deep learning" OR "neural networks") AND ("generalization" OR "cross-domain" OR "out-of-distribution").

1.5.2 Literature selection criteria

We applied a two-tiered filtering process to delineate the literature selection criteria of this survey:

Tier 1: Core survey corpus (the subject of analysis). To capture the state-of-the-art (SOTA) landscape, papers included in the detailed analysis and taxonomy must meet the following criteria:

- *Time window*: Published between January 2020 and early 2025.
- *Modality*: Focus on unimodal DF detection in image, video, or audio modalities.
- *Methodology*: Employ deep learning-based techniques for DF detection.
- *Venue*: Published in peer-reviewed conferences or journals in AI, computer vision, or security domains (e.g., CVPR, AAAI, ICML, ICLR, ICCV, ECCV, NeurIPS, ICASSP, IEEE TIFS, IEEE T-PAMI).*Tier 2: Foundational Reference (background context).* We

also include seminal and foundational works that provide essential context for understanding the evolution of DF detection techniques. These works may fall outside the primary time window or modality focus but are critical for comprehending the field's development. The inclusion criteria for this tier are:

- *Foundation datasets*: Papers introducing widely adopted DF datasets (i.e., DF-TIMIT (Korshunov 2018), FaceForensics++ (Rossler et al. 2019), DFDC (Dolhansky et al. 2019)).
- *Foundation methods*: Papers announcing pioneering DF detection methods that have significantly influenced subsequent research in audio DF detection (i.e., (Yang et al. 2018; Alzantot et al. 2019), and (Saleem et al. 2019)). The purpose is to help readers find seminal works that laid the groundwork for current approaches.

1.5.3 Exclusion criteria

We excluded studies that: (1) Focus exclusively on generation without detection; (2) Rely on active defense (watermarking) or pure manual forensics; or (3) Lack results or evaluations.

1.6 Survey structure

The rest of this survey is structured as follows: Section 2 presents common benchmarks, datasets, and metrics. We review unimodal DF detection methods across image, video, and audio modalities and analyze inter-modality relationships in Section 3. Section 4 provides a comprehensive review of generalization and robustness techniques. We present out comprehensive empirical evaluation in Section 5, which first verifies reported results in original papers, then evaluates generalization and robustness capabilities of SoTA methods. Section 6 discusses current challenges and future research directions. Finally, we conclude the survey in Section 7. An overview of our survey structure is illustrated in Figure 1. To enhance readability, Table 3 provides a comprehensive list of abbreviations used throughout this survey, organized by category.

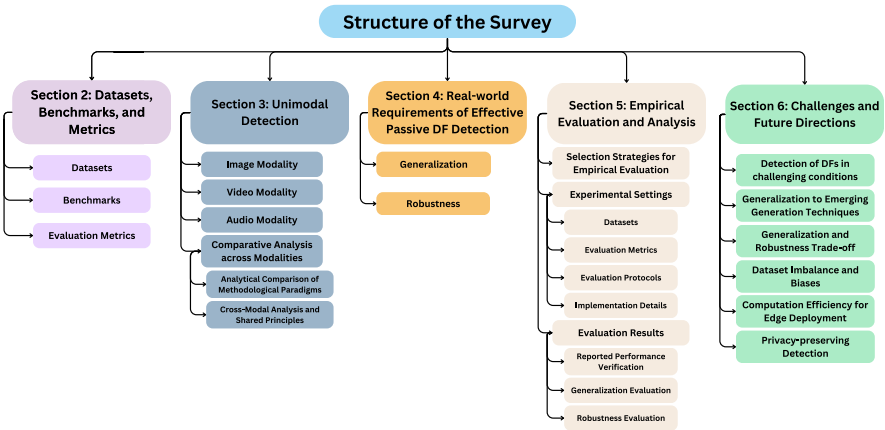


Fig. 1 Overview of our survey structure

Table 3 List of abbreviations

Category	Abbreviation	Full term
Core terms		
	DF	Deepfake
	GM	Generative model
	GAN	Generative adversarial network
	DM	Diffusion model
	LLM	Large language model
	AIGC	AI-Generated content
	SoTA	State-of-the-Art
Manipulation types		
	FS	Face swap
	FR	Face reenactment
	FE	Face editing/Face enhancement
	ES	Entire synthesis
	VC	Voice conversion
	TTS	Text-to-speech
Neural network architectures		
	CNN	Convolutional neural network
	RNN	Recurrent neural network
	LSTM	Long short-term memory
	ViT	Vision transformer
	GNN	Graph neural network
	TCN	Temporal convolutional network
Learning paradigms		
	DL	Deep learning
	SSL	Self-supervised learning
	CL	Contrastive learning
	UCL	Unsupervised contrastive learning
	E2E	End-to-End
	MAE	Masked autoencoder
Evaluation metrics		
	ACC	Accuracy
	AUC	Area under the curve
	EER	Equal error rate
	FID	Fréchet inception distance
	SSIM	Structural similarity index
	PSNR	Peak signal-to-noise ratio
Adversarial attacks		
	PGD	Projected gradient descent
	HSJA	HopSkipJumpAttack
	FGSM	Fast gradient sign method
	C&W	Carlini & Wagner attack

2 Datasets, benchmarks, and metrics

This section provides an overview of commonly used datasets (Section 2.1), benchmarks (Section 2.2), and evaluation metrics (Section 2.3) in passive DF detection across image, video, and audio modalities. Understanding these resources is essential for evaluating and

comparing detection methods effectively.

2.1 Datasets

The DF detection field has produced extensive datasets across visual and audio modalities, which are summarized in Table 4. Visual modality datasets dominate with over 20 publicly available resources (Dang et al. 2020; He et al. 2021; Nguyen et al. 2021; Wang et al. 2023; Yan et al. 2024; Yang et al. 2019; Korshunov 2018; Rossler et al. 2019; Dolhansky et al. 2019; Li et al. 2020; Jiang et al. 2020; Zi et al. 2020; He et al. 2021; Zhou et al. 2021; Kwon et al. 2021; Narayan et al. 2022, 2023). Audio modality datasets are fewer, with 11 notable datasets (Wang et al. 2020; Liu et al. 2023; Frank and Schönherr 2021; Yi et al. 2022, 2023; Li et al. 2024; Müller et al. 2022; Yaroshchuk et al. 2023; Zhang et al. 2022; Müller et al. 2024; Yan et al. 2024).

In the visual domain, FaceForensics++ (FF++) (Rossler et al. 2019) is the most widely used training dataset, containing over 1,000 videos manipulated with four different techniques: DeepFakes, Face2Face, FaceSwap, and NeuralTextures. Building on FF++, subsequent datasets expanded in scale and diversity. DFDC (Dolhansky et al. 2019) increased volume with 4,113 fake videos, while CelebDF (Li et al. 2020) focused on improved visual quality with 5,689 celebrity DFs. ForgeryNet (He et al. 2021) further scaled to 2.9 million images and 220,000 videos across 15 manipulation methods. Recent datasets incorporate Diffusion Model (DM) generations to generate DFs. For example, DF40 (Yan et al. 2024) which includes 40 distinct synthesis models spanning both GANs and DMs and DiffusionForensics (Wang et al. 2023) specifically targets DMs with 6 different architectures. The audio modality shows similar evolution patterns. ASVSpoof challenges (Wang et al. 2020; Liu et al. 2023) provided first audio DF datasets with hundreds of thousands of samples targeting voice conversion and text-to-speech attacks. WaveFake (Frank and Schönherr 2021) introduced a large-scale dataset with 117,985 samples generated by various SOTA models. More recent datasets like ADD (Yi et al. 2023), MLAAD (Müller et al. 2024), and VoiceWukong (Yan et al. 2024) have expanded the diversity of synthesis techniques and languages. They cover up to 38 languages and 50 synthesis models.

However, these datasets exhibit significant limitations that hinder the advancement of real-world DF detection methods. The first issue is severe class imbalance, where fake samples often outnumber real ones, leading to biased training and poor generalization (Layton et al. 2024). For instance, DFDC (Dolhansky et al. 2019) has a 3:1 ratio of fake to real videos, while CelebDF (Li et al. 2020) has nearly 5 times more fake videos. Only VoiceWukong (Yan et al. 2024) maintains a balanced distribution between real and fake samples. The second issue is the lack of challenging scenarios. Most datasets contain relatively clean, single-face manipulations under controlled conditions. Only a few datasets like OpenForensics (Nguyen et al. 2021), DF-Platter (Narayan et al. 2023), and DeeShy (Narayan et al. 2022) address challenging scenarios. These include multiple fake faces, low-quality inputs, or face occlusion. Third, many datasets focus on specific generation techniques (e.g., GANs) and do not reflect the latest advancements in GMs (e.g., DMs). While newer datasets like DF40 (Yan et al. 2024) and DiffusionForensics (Wang et al. 2023) have started to include diffusion-based DFs, most existing datasets still lack this diversity. Finally, the most concerning is the complete absence of adversarial robustness in testing in current datasets. None

of the existing datasets incorporate adversarially perturbed DFs designed to evade detection, which is a critical real-world threat (Corvi et al. 2023; Hou et al. 2023).

2.2 Benchmarks

DeepfakeBench (Yan et al. 2023) is the first comprehensive and public benchmark for DF detection in image and video modalities. It offers an integrated framework featuring 34 detectors and 10 datasets, alongside standardized data preprocessing and evaluation protocols. Regarding the audio modality, XMAD-Bench (Ciobanu et al. 2025) is the first comprehensive benchmark which support 12 SoTA detectors and 7 languages. However, these benchmarks do not cover datasets for challenging scenarios, such as multiple fake faces, face occlusions, partial fake, and low quality. Recently, (Li et al. 2023) introduces a continual benchmark that simulates a diverse stream of DFs sequentially over time, challenging detection models to learn new fake types without forgetting previously encountered ones. Despite its innovative approach to testing detector robustness against catastrophic forgetting, CDDB has received limited adoption in subsequent research.

2.3 Evaluation metrics

Standard evaluation metrics for DF detection include accuracy (ACC), precision, recall, F1-score, area under the receiver operating characteristic curve (AUC-ROC), and equal error rate (EER). Accuracy measures the overall correctness of predictions but can be misleading in imbalanced datasets. Precision quantifies the proportion of true positive detections among all positive predictions, while recall measures the proportion of actual positives correctly identified. The F1-score provides a harmonic mean of precision and recall, balancing both metrics. AUC-ROC evaluates the trade-off between true positive and false positive rates across different thresholds, providing a comprehensive measure of model performance. EER represents the point where false positive and false negative rates are equal, commonly used in audio DF detectors. For robustness evaluation against adversarial attacks, the attack success rate (ASR) metric is commonly used.

3 Unimodal detection

This section provides a comprehensive overview of unimodal detection approaches to identify manipulated content in each modality. For each modality, we categorize existing methods based on their underlying detection principles and the provenance of the features exploited. Figure 2 illustrates the taxonomy of unimodal DF detection approaches. In the following subsections, we first provide a comprehensive review of SoTA approaches for each modality (Section 3.1 for image, 3.2 for video, 3.3 for audio), then discuss their strengths, limitations and inter-modality relationships in Section 3.4.

3.1 Image modality

We classify DF image detection methods into four categories: Forensic-based, Data-driven, Fingerprint-based, and Hybrid. This classification distinguishes methods based on their

Table 4 A Summarization of datasets for each modality in passive df detection approaches

	Name	Year	Samples		Gen.		SM	Mani.Tech.	Mult.Faces	LowQual.	FaceOccl.	Lang.	PartialFake
			Real	Fake	GANs	DMs							
Continued on next page													
Image	DFFD (Dang et al. 2020)	2019	58,703	240,336	✓	×	7	FS, FR, FE	×	×	×	-	×
	ForgeryNet (He et al. 2021)	2021	1,438,201	1,457,861	✓	×	15	FS, FR, FE	×	×	×	-	×
	OpenForensics (Nguyen et al. 2021)	2021	160,467	173,660	✓	×	1	FS	✓	×	✓	-	×
	DiffusionForensics (Wang et al. 2023)	2023	134,000	137,200	×	✓	9	ES	×	×	×	-	×
	DiffusionFace (Chen et al. 2024)	2024	30,000	600,000	×	✓	11	FS, ES, FE	×	×	×	-	×
	DF40 (Yan et al. 2024)	2024	1590	1 M	✓	✓	40	ES, FE	×	×	×	-	×
Video	UADFV (Yang et al. 2019)	2018	49	49	✓	×	1	FS	×	×	×	-	×
	DF-TIMIT (Korshunov 2018)	2018	320	640	✓	×	2	FS	×	✓	×	-	×
	DFFD (Dang et al. 2020)	2019	1,000	3,000	✓	×	7	FS, FR, FE	×	×	×	-	×
	FF++ (Rossler et al. 2019)	2019	1,000	4,000	✓	×	4	FS, FR	×	×	×	-	×
	DFDC (Dolhansky et al. 2019)	2019	1,131	4,113	✓	×	8	-	×	×	×	-	×
	Celeb-DF (Li et al. 2020)	2020	590	5,639	✓	×	1	FS, FR	×	×	×	-	×
	DF−1.0 (Jiang et al. 2020)	2020	50,000	10,000	✓	×	1	FS	×	×	×	-	×
	Wild-DF (Zi et al. 2020)	2021	3,805	3,509	✓	×	1	–	×	×	×	–	×
	ForgeryNet (He et al. 2021)	2021	99,630	121,617	✓	×	15	FS, FR, FE	×	×	×	–	×
	FFIW (Zhou et al. 2021)	2021	10,000	10,000	✓	×	3	FS	×	×	×	–	×
	KoDF (Kwon et al. 2021)	2021	62,166	175.776	✓	×	6	FS, FR	×	×	×	–	×
	DeeShy (Narayan et al. 2022)	2022	100	5,040	✓	×	3	FS, FR	×	×	✓	–	×
	DF-Platter (Narayan et al. 2023)	2023	133,260	132,496	✓	×	3	FS	✓	✓	✓	–	×
	DF40 (Yan et al. 2024)	2024	1590	0.1M	✓	✓	40	FS, FR	×	×	×	–	×
Audio	ASVspoof 2019 (Wang et al. 2020)	2019	41,913	300,678	–	–	–	VC, TTS	–	×	–	en	×
	ASVspoof 2021 (Liu et al. 2023)	2021	22,617	589,212	–	–	–	VC, TTS	–	×	–	en	×
	WaveFake (Frank and Schönherr 2021)	2021	18,100	117,985	✓	×	6	VC	–	×	–	en, jp	×
	ADD 2022 (Yi et al. 2022)	2022	36,953	123,932	–	–	–	VC	–	✓	–	ch	✓
	ADD 2023 (Yi et al. 2023)	2023	172,819	113,042	–	–	–	VC	–	✓	–	ch	✓
	ITW (Müller et al. 2022)	2022	17,000	14,000	–	–	–	VC	–	×	–	en	×

Table 4 (continued)

Name	Year	Samples		Gen.		SM	Mani.Tech.	Mult.Faces	LowQual.	FaceOccl.	Lang.	PartialFake
		Real	Fake	GANs	DMs							
ODSS (Yaroshchuk et al. 2023)	2023	11,032	18,993	✓	×	2	TTS	–	×	–	en, es, de	×
PS (Zhang et al. 2022)	2023	12,483	121,461	–	–	–	VC, TTS	–	×	–	en	✓
CD-ADD (Li et al. 2024)	2024	28,212	120,459	✓	✓	5	VC, TTS	–	×	–	en	×
MLAAD (Müller et al. 2024)	2024	20,000	134,000	✓	✓	82	TTS	–	×	–	many (38)	×
VoiceWukong (Yan et al. 2024)	2024	413,400	413,400	✓	✓	34	VC, TTS	–	✓	–	en, ch	×

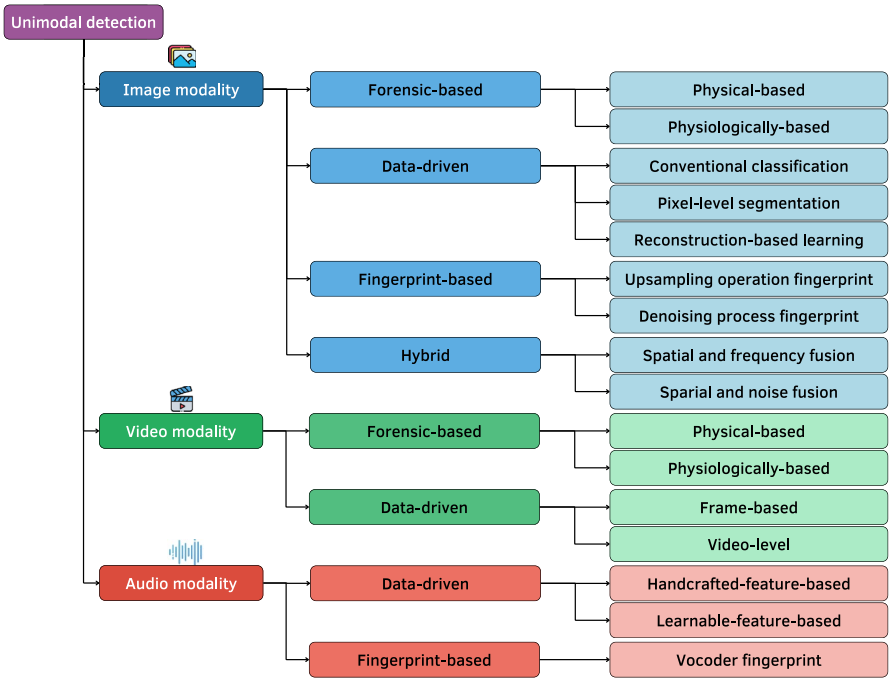


Fig. 2 Taxonomy of unimodal DF detection approaches

input hypotheses and feature extraction strategies. *Forensic-based methods* operate on the premise that generators fail to replicate physical-world constraints. *Data-driven methods* operate on the premise that massive datasets allow deep networks to find statistical anomalies invisible to human design. *Fingerprint-based methods* operate on the premise that the generative architecture itself (e.g., GAN or DM) imprints a specific signature regardless of the image content. *Hybrid methods* explicitly fuse these feature spaces to leverage complementary cues. Figure 3 illustrates four main categories in the image modality.

Forensic-based methods. Methods in this group detect DFs based on predefined rules related to differences in physical-based and physiologically-based signals between real and fake images. For example, the inconsistencies between left and right eyes or the unnaturalness of lighting conditions in fake images. The key insight behind these approaches is that generators cannot perfectly replicate the complex physical properties and biological consistencies inherent in natural image formation. Some studies reveal that natural images exhibit consistent relationships between luminance and chrominance components (Chen et al. 2021; Talib et al. 2025). Thus, analyzing multiple color spaces can provide a more comprehensive view of potential manipulation traces. Other works use cross-color spatial co-occurrence matrices (Qiao et al. 2023) or Wasserstein distance (Amin et al. 2024) to analyze statistical relationships between color channels of real and fake images. Biological-based inconsistencies are also analyzed to identify DFs, such as the differences between left and right eyes (Hu et al. 2021; Wang et al. 2022) and the discrepancies between the internal face region and the surrounding context (Nirkin et al. 2021).

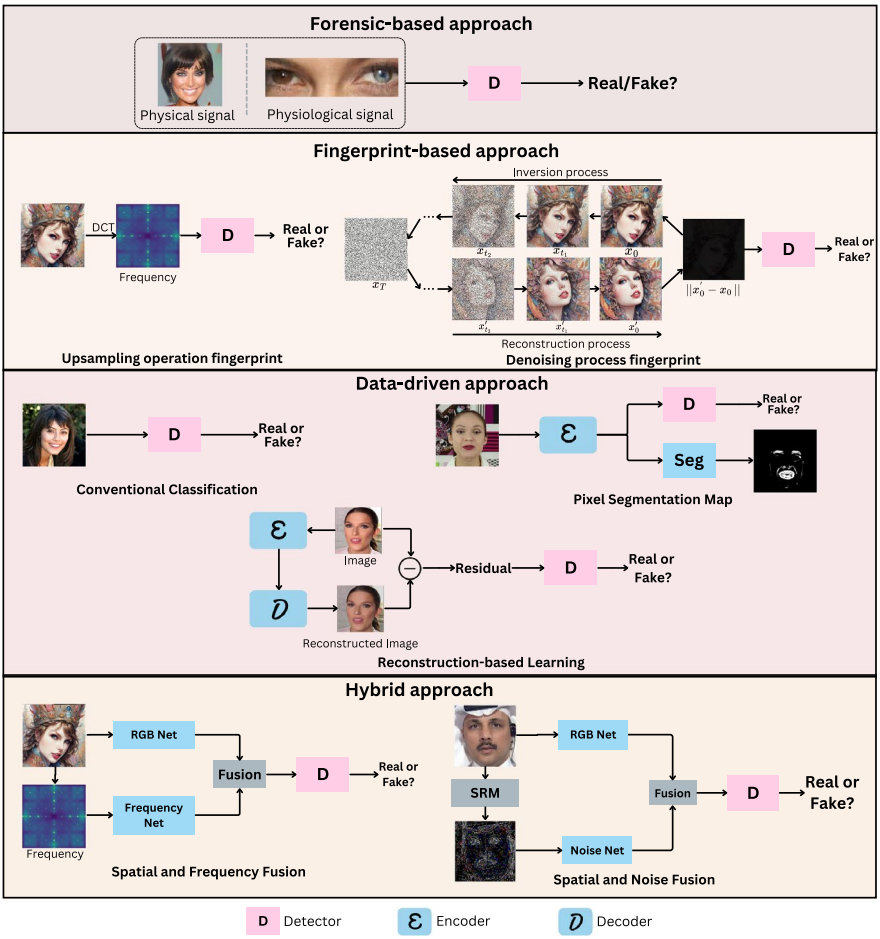


Fig. 3 Taxonomy and architectural overview of passive image DF detection. The diagram categorizes methods into Forensic-based, Data-driven (including reconstruction and segmentation), Fingerprint-based, and Hybrid approaches, illustrating the distinct input-processing pipelines used to identify spatial artifacts

Data-driven methods. Instead of relying on predefined rules, these methods leverage deep learning (DL) to automatically learn discriminative features from large-scale datasets. Based on how they formulate the detection problem, data-driven approaches can be categorized into three main groups.

① *Conventional classification* methods cast DF detection as a binary classification problem, where the model learns to classify an input image as real or fake (Yang et al. 2021; Sha et al. 2023). However, simple binary classification is often suboptimal given the subtle, localized differences between real and fake images (Zhao et al. 2021; Han et al. 2023). Zhao et al. (2021) (Zhao et al. 2021) reformulated as a fine-grained classification task that enables the detector to focus on local regions. Meanwhile Han et al. (2023) (Han et al. 2023) reformulated as multi-task learning by designing a network that can detect multiple types of DF face images simultaneously.

② *Pixel-level segmentation* methods formulate detection as a dense prediction task that generates localization maps to identify manipulated regions at the pixel level. By predicting manipulation masks, these methods not only determine whether the input is real or fake but also pinpoint which image regions have been manipulated. This additional localization capability provides more interpretable results and can help understand the manipulation techniques used. These methods localize the manipulated area in a fully supervised setting. They use either an encoder-decoder architecture (Huang et al. 2022; Wang et al. 2022; Mazaheri and Roy-Chowdhury 2022; Katamneni and Rattani 2024) or an attention mechanism (Das et al. 2022; Guo et al. 2023; Hong et al. 2024). However, these require dense pixel-wise ground-truth masks, which might not always be available. Tantaru et al. (2024) (Tantaru et al. 2024) overcomes this limitation by applying the GradCAM explainability technique to the activations produced by a classification network to highlight the regions most predictive of the fake class.

③ *Reconstruction-based learning* methods formulate detection through the lens of image reconstruction. These approaches typically attempt to reconstruct the input image using an encoder-decoder architecture and analyze the reconstruction errors. The key principle is to focus on learning what real images should look like through reconstruction, and the reconstruction of fake images is different from that of real images. However, Shi et al. (2023) (Shi et al. 2023) noted that single reconstruction-based methods (He et al. 2021; Cao et al. 2022; Liu et al. 2023; Guo et al. 2024) lack feature depth. To address this, they proposed a double-head module combining discrepancy-guided encoding and dual reconstruction to improve performance.

Fingerprint-based methods. Fingerprint-based methods for DF detection explore the unique patterns or artifacts unintentionally embedded into the generated images by the architectural design and training process of GMs. The key insight is that different GMs leave distinct *fingerprints* in their outputs due to their architectural design and training process. For GAN-generated images, these methods focus on artifacts introduced by the upsampling operations commonly used in generator architectures. Several studies have shown that the frequency spectrum of GAN-generated images often contains distinctive checkerboard patterns, particularly in the middle and high-frequency bands (Frank et al. 2020; Qian et al. 2020; Dzanic et al. 2020; Liu et al. 2020; Tan et al. 2024). Regarding images generated by DMs, the iterative denoising process of DMs tends to leave characteristic traces that can be detected through careful analysis (Wang et al. 2023; Ma et al. 2023). Recently, Tan et al. (2024) (Tan et al. 2024) explores how upsampling operations in GANs and DMs create distinctive patterns in how neighboring pixels relate to each other, while the work (Tan et al. 2024) applies convolutional layers to the phase and amplitude spectra of the network's internal feature maps rather than solely analyzing the frequency artifacts of the input image.

Hybrid methods. Detection approaches that solely rely on spatial domain information have shown to be highly susceptible to variations in dataset quality and post-processing operations (Qian et al. 2020; Wang et al. 2020). To address this limitation, hybrid approaches aim to leverage complementary information from different domains to enhance detection performance. The key insight is that DFs often leave traces across multiple feature spaces. Combining these cues can provide more robust detection than relying on a single domain. Based on the types of features being fused, hybrid methods can be categorized into two main groups.

① *Spatial and frequency fusion* methods combine features from the spatial domain (capturing visual content and local artifacts) with frequency domain information (revealing generation artifacts in the spectral space). The spatial stream typically employs convolutional networks to extract content-level features, while the frequency stream analyzes spectral patterns through discrete cosine transform (DCT) or discrete Fourier transform (DFT) (Miao et al. 2023, 2022; Wang et al. 2022, 2023; Luo et al. 2021). This dual-stream architecture helps capture both semantic inconsistencies and subtle frequency-domain artifacts introduced during the generation process.

② *Spatial and noise fusion* methods integrate standard RGB features with noise patterns extracted through SRM noise filters (Fridrich and Kodovsky 2012). These approaches recognize that DF generation often leaves distinctive noise patterns that are different from those found in authentic images. By analyzing both content features and noise residuals, these methods can better distinguish between natural imaging noise patterns and synthetic artifacts (Kong et al. 2022, 2022; Shuai et al. 2023). Instead of relying on handcrafted SRM noise filters, TruFor (Guillaro et al. 2023) trains Noiseprint++ extractor in a self-supervised manner using contrastive learning to capture stronger noise traces.

3.2 Video modality

Compare with image modality. Video-based DF detection presents unique challenges (Vahdati et al. 2024). On the one hand, approaches developed for image detection can be naturally extended to videos by applying them to individual frames. Forensic-based methods can analyze physical or physiological inconsistencies in each frame, while data-driven approaches can classify frames independently. The final decision can then be obtained by aggregating frame-level predictions through voting or averaging mechanisms. However, this frame-by-frame analysis fails to capture a crucial aspect of DF videos: temporal coherence. Manipulated videos often exhibit subtle temporal inconsistencies. These include unnatural head movements, inconsistent facial expressions across frames, and flickering artifacts in manipulated regions. These temporal artifacts cannot be detected by examining frames in isolation, making pure image-based approaches insufficient for robust video DF detection. Therefore, it is imperative to develop dedicated video-level approaches that explicitly model temporal relationships. In this survey, we categorize detection approaches in the video modality into two primary groups: (i) *forensic-based* and (ii) *data-driven*, which are illustrated in Figure 4.

Forensic-based methods primarily extend insights from image-modality forensic analysis to the video modality by examining physical and biological inconsistencies frame by frame. Xia et al. (2022) (Xia et al. 2022) analyzes multiple color channels, while Huda et al. (2024) (Huda et al. 2024) uses multiple texture feature descriptors to capture surface inconsistencies per frame. Other studies analyze physiological signals that should remain consistent in authentic videos but may be imperfectly replicated in DFs, such as blood flow patterns through skin color changes (Jeon et al. 2022), facial movements (Sun et al. 2021; Li et al. 2023), or the difference between front and side face images (Li et al. 2023). Advanced methods (Sun et al. 2021; Li et al. 2023) utilize Recurrent Neural Network to model the temporal characteristics on the embedded feature sequences.

Data-driven approaches. Based on how temporal information is processed, these methods can be categorized into two main approaches: Frame-level and Video-level.

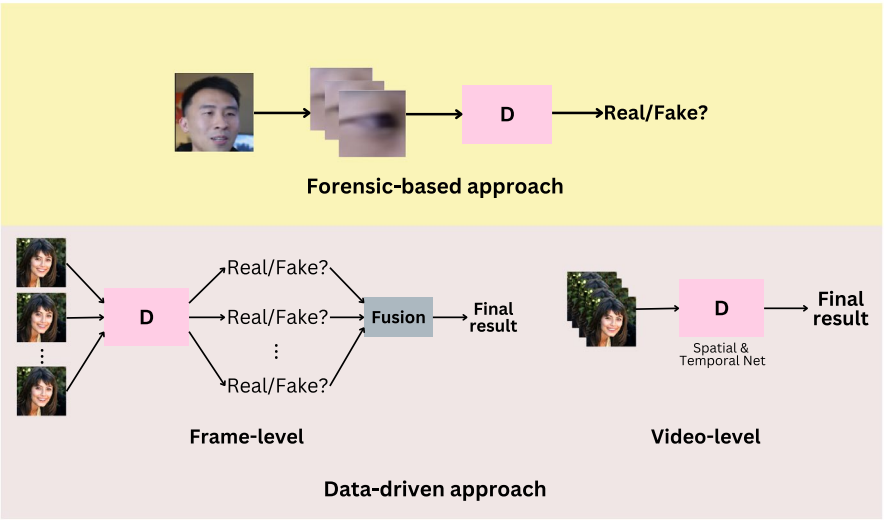


Fig. 4 Taxonomy and architectural overview of passive video DF detection. The diagram contrasts Frame-level approaches, which aggregate single-image predictions, with Video-level approaches that utilize spatiotemporal networks (e.g., 3D CNNs, Transformers) to capture temporal inconsistencies and identity flickering

① *Frame-level methods* essentially apply image-modality DL techniques to individual frames, treating the video as a sequence of independent images. These methods are typically based on CNN architectures to classify each frame as real or fake, then aggregate these frame-level predictions through techniques like majority voting or temporal averaging to make a final video-level decision (Wang and Chow 2023; Ciamarra et al. 2024; Bonettini et al. 2021). To capture a broad set of forgery clues (e.g., blending ghosts, skin tone inconsistencies, tooth details, stitching seams), Ba et al. (2024) (Ba et al. 2024) proposes a local disentanglement module to extract multiple local representations and then fuses them into a global semantic-rich feature. Rather than relying on per-frame, some works (Gu et al. 2021, 2022a, b, c) have focused on capturing local inconsistency within densely sampled video snippets, where the entire video is densely divided into multiple snippets. While straightforward to implement, these frame-based approaches may miss temporal inconsistencies that are only visible when analyzing frame sequences.

② *Video-level methods* explicitly model temporal relationships by utilizing architectures designed to capture both spatial and temporal information. Early attempts combined convolution-based networks and recurrent-based networks to extract global spatio-temporal features (Montserrat et al. 2020; Saikia et al. 2022); however, this approach has been demonstrated to be less effective (Zhao et al. 2023). Recent approaches have utilized advanced architectures that can capture long-range dependencies and inter-frame inconsistencies. These include transformers (Zhao et al. 2023; Coccomini et al. 2022; Zhao et al. 2022; Zhang et al. 2022), 3D CNN (Zhang et al. 2021), temporal convolution networks (Zheng et al. 2021) and attention mechanism (Hu et al. 2024). Another line of work aims at identifying the identity-related inconsistencies in DF videos. These approaches recognize that while DFs may maintain visual quality in individual frames, they often struggle to consistently preserve identity characteristics across an entire video sequence. These methods typically

employ two main strategies. The first strategy is to utilize pre-trained face recognition to extract identity embeddings and analyze how these embeddings evolve over time to detect unnatural variations (Agarwal et al. 2020; Huang et al. 2023). The second strategy focuses on capturing spatio-temporal identity features directly through end-to-end training through 3D face reconstruction or identity-specific transformer (Cozzolino et al. 2021; Liu et al. 2023; Dong et al. 2022). Recently, MinTime (Coccomini et al. 2024) aims to address a crucial limitation in previous identity-driven approaches that struggle with videos containing multiple people. The authors introduce an identity-aware attention mechanism that applies masking to process each identity in the video independently. This enables the detector to handle multiple identities in the video.

3.3 Audio modality

Compare with image modality. Methodologies developed for visual media cannot be directly applied to detect DF audio (Lee et al. 2024). Image DF detection typically relies on visual artifacts such as inconsistent lighting, unnatural eye reflections, or imperfect facial blending. These artifacts can be analyzed using CNNs trained on pixel-level data. In contrast, audio signals exist primarily as 1D temporal waveforms or 2D time-frequency representations. In these representations, artifacts manifest as spectral anomalies, inconsistent phase coherence, or irregular prosodic patterns. For example, synthetic voices often exhibit unnatural pauses, overly consistent pitch modulation, or artifacts in high-frequency bands. These are imperceptible to humans but detectable via mel-spectrogram analysis. Given this distinct nature of audio signals, current *data-driven approaches* differ primarily in how they extract discriminative acoustic features. To clarify the intersection of feature types noted in recent literature, we categorize these methods based on the source of the feature definition. Handcrafted-feature-based methods rely on expert-defined signal processing (e.g., MFCC, CQT) to fix the feature space prior to classification. In contrast, learnable feature-based methods leverage neural networks to extract representations directly from raw audio or spectrograms in an end-to-end manner, prioritizing the automatic discovery of complex temporal patterns. Distinct from general acoustic anomalies, *fingerprint-based approaches* explore artifacts left by speech synthesis models, particularly vocoders. Figure 5 illustrates these approaches.

Data-driven methods. Based on the front-end feature extraction techniques employed, data-driven methods can be classified into two main categories.

① *Handcrafted-feature-based methods* utilize expert-based audio processing techniques as front-end to convert the audio waveform into predefined acoustic features. These features fall into two main categories - physical and perceptual - each capturing different aspects that help distinguish between authentic and synthetic speech (Li et al. 2022). Mel Frequency Cepstral Coefficients (MFCC) capture the spectral envelope of speech. Synthetic audio often exhibits unnatural patterns in these features due to imperfect modeling of vocal tract characteristics (Alzantot et al. 2019). Constant Q transform (CQT) and spectrogram representations reveal frequency distributions and power spectrum patterns that may be distorted in DF audio (Yang et al. 2018; Xue et al. 2022; Kwak et al. 2021; Wani et al. 2024). In contrast, perceptual features are designed to capture characteristics that align with human auditory perception and natural speech properties. Techniques like Jitter and Shimmer measure frequency and amplitude variations, respectively, capturing micro-perturbations that are naturally present in human voice but often missing or incorrectly reproduced in

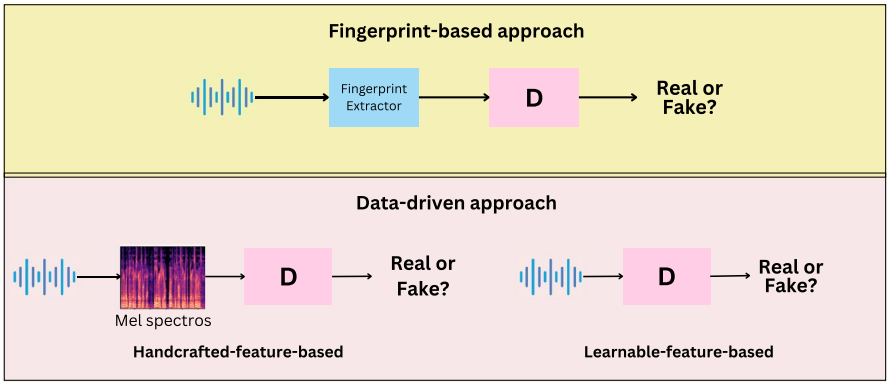


Fig. 5 Taxonomy and architectural overview of passive audio DF detection. The figure differentiates between Data-driven methods, which analyze handcrafted (e.g., MFCC) or learnable acoustic features, and Fingerprint-based methods that target specific vocoder artifacts to distinguish synthetic speech from natural audio

synthetic speech (Gao et al. 2021). Chromagram analysis examines pitch-related features, helping identify unnatural pitch progressions or tonal qualities that may occur in DF audio (Saleem et al. 2019). These extracted features provide rich information for the classifier to distinguish between real and fake audios.

② *Learnable feature-based methods* leverage NNs to automatically extract discriminative features directly from raw audio in an end-to-end (E2E) manner. These methods design specialized network architectures that learn optimal feature representations through supervised learning, such as the spectro-temporal graph attention network (Jung et al. 2022; Huang et al. 2023), Inception-style architecture (Towards end-to-end synthetic speech detection 2021), RawNet-based architecture (Tak et al. 2021), and Conformer-based architecture (Shin et al. 2024; Rosello et al. 2023). Taking the advantage of Mamba (Gu and Dao 2023), Chen et al. (2024) (Chen et al. 2024) proposes a novel E2E bidirectional state space model for audio DF detection to capture more long-range relationships. However, RawBMamba lacks a mechanism to capture localized artifacts that manifest in specific spectro-temporal regions. To address this, BiCrossMamba-ST (El Kheir et al. 2025) utilizes a 2D-attention map to dynamically identify these critical regions within the feature space. Additionally, a mutual cross-attention module is introduced to allow the spectral and temporal branches to exchange information, enhancing the model's ability to detect subtle forgery traces. El Kheir et al. (2025) (El Kheir et al. 2025) combines a self-supervised model with Mamba to effectively extract rich features and capture long-range discriminative patterns in speech. Yang et al. (2024) (Yang et al. 2024) introduces a multi-view feature fusion approach that concatenates handcrafted and learnable features to capture a broad range of audio features. WhiADD (Cui et al. 2025) enhances audio DF detection by fine-tuning the Whisper decoder with full-transcript semantic prompts, effectively bridging the gap between low-level acoustic artifacts and high-level linguistic coherence. For resource-constrained applications, Chakravarty and Dua (2024) (Chakravarty and Dua 2024) propose a lightweight feature extractor using linear discriminant analysis to reduce a 2048-dimensional feature vector to a single, highly discriminative feature. Recently, Mixture-of-Experts architecture has been explored by Negroni et al. (2025) (Negroni et al. 2025) to dynamically weight con-

tributions from multiple specialized expert detectors. By leveraging expert specialization across diverse datasets, this method effectively enhances accuracy performance of audio DF detection.

Inspired by advancements in speech Self-Supervised Learning (SSL), several works (Wang and Yamagishi 2024; Tran et al. 2024; Guo et al. 2024) leverage (SSL speech models like Wav2vec2 and XLS-R as front-end feature extractors for DF detection. These models, pre-trained on large amounts of unlabeled speech data, demonstrate remarkable capacity for capturing subtle acoustic characteristics across diverse speaking conditions. However, naively fine-tuning these large pre-trained models for DF detection presents challenges. These include overfitting with limited training data and computational overhead (Martín-Doñas et al. 2024; Wu et al. 2024). To address these concerns, Martín-Doñas et al. (2024) (Martín-Doñas et al. 2024) a lightweight downstream classifier with minimal trainable parameters to preserve the generalized audio representations of the SSL model while Pan et al. (2024) (Pan et al. 2024) selectively fine-tunes only early and middle layers to reduce computational requirements. To capture both local and global artifacts in raw waveforms, Tran et al. (2025) (Tran et al. 2025) designs a novel front-end architecture that integrates a gating mechanism with a pre-trained XLS-R foundation model. Beyond traditional supervised learning paradigms, some researchers have explored alternative training strategies, including re-synthesizing real samples using various neural vocoders (Doan and Nguyen-Vu 2024) or modeling only genuine speech representations and classifying any sample falling outside these established boundaries as fake (Kim et al. 2024).

Fingerprint-based methods. Similar to the image modality, these methods recognize that different vocoder architectures produce characteristic artifacts in the frequency spectrum, phase patterns, or temporal structure of the generated audio. The core idea is to identify and extract these vocoder-specific "fingerprints" that may be detectable through signal analysis. For instance, neural vocoders often struggle to reproduce certain aspects of natural speech, such as phase coherence across frequency bands or precise harmonic structures. Particularly, these methods have explored that the spectrograms of original audio have more natural and consistent high-frequency components and more natural harmonic structures compared to synthetic audio (Yan et al. 2022; Sun et al. 2023).

3.4 Comparative analysis across modalities

3.4.1 Analytical comparison of methodological paradigms

Handcrafted vs. end-to-end representation learning. There exists a distintion between visual and audio detection paradigms regarding feature extraction.

- *Visual modality (Image/Video)*: The predominant trend in visual DF detection leans towards end-to-end representation learning. The spatial correlations in pixels are effectively captured by neural networks (e.g., CNNs or ViTs) without the need for handcrafted feature extraction. Forensic-based methods are increasingly marginalized due to their inability to scale with modern high-fidelity generators.
- *Audio modality*: In contrast, audio DF detection retains a strong reliance on handcrafted features (e.g., MFCC, CQT, LFCC). Unlike the pixel domain, the raw audio waveform is high-dimensional and less semantically structured. Consequently, audio detectors

frequently require "front-end" signal processing to transform waveforms into spectrograms or cepstral features before classification.

This divergence suggests that visual detectors have successfully automated feature discovery. However, audio detection still benefits significantly from domain-specific signal processing knowledge to filter out noise and focus on vocoder artifacts.

The role of temporal dynamics. Both video and audio modalities require modeling time, but they treat temporal information differently.

- *Video*: Temporal inconsistency is typically treated as a secondary factor to spatial features. Most video detectors first extract strong frame-level spatial features and then use lightweight aggregators (LSTMs, 3D-CNNs) to detect flickering (Cozzolino et al. 2021; Wang et al. 2023). The primary signal remains spatial. Temporal analysis often serves to check for consistency between frames.
- *Audio*: Temporal coherence is the primary signal. A single millisecond of audio is meaningless; artifacts exist only in sequence. Therefore, audio detectors (e.g., RawNet (Kawa et al. 2022), Conformer (Shin et al. 2024)) must model long-range dependencies intrinsically, unlike video detectors which can often perform reasonably well using simple frame averaging.

This explains why detection methods in the image domain can be easily adapted to the video domain, whereas applying image-based architectures to audio spectrograms often fails to capture the necessary phase continuity and prosodic patterns unique to speech.

The interpretability vs. performance trade-off. Across all modalities, there is a consistent trade-off between interpretability and raw detection performance.

- *Forensic-based methods*: These methods offer high interpretability by relying on predefined physical or physiological inconsistencies. They provide clear rationales for classification decisions. However, their performance is often lower than data-driven approaches. This gap widens as generators improve and better mimic real-world properties.
- *Fingerprint-based methods*: Whether detecting checkerboard upsampling artifacts in images or vocoder fingerprints in audio, these methods offer high interpretability. They provide a clear rationale for classification. However, they are highly fragile; a change in the generator architecture (e.g., moving from GAN to Diffusion, or WaveNet to HiFi-GAN) often renders specific fingerprint detectors obsolete.
- *Data-driven methods*: These offer superior generalization to unknown datasets by learning abstract boundaries. However, they act as "black boxes," providing little insight into the specific artifacts driving decisions. This lack of interpretability can be problematic in high-stakes scenarios where understanding the rationale behind a detection is crucial.

Tables 5-7 summarize the key ideas, strengths, and limitations of unimodal approaches across modalities See Table 6.

3.4.2 Cross-modal analysis and shared principles

While Section 3.4.1 highlights the unique strengths and limitations of passive DF detection methods, there are notable commonalities and shared principles across modalities that can inform future research directions.

Unified artifact across modalities. Despite fundamental differences in data representation, DF artifacts across all three modalities can be categorized into a unified taxonomy based on their origin:

- *Generation process artifacts* arise from architectural limitations of GMs. In images, upsampling operations in GANs produce checkerboard patterns detectable in the frequency domain (Frank et al. 2020; Tan et al. 2024; Liu et al. 2020; Dzanic et al. 2020; Qian et al. 2020). In videos, frame-by-frame generation creates temporal flickering and inter-frame inconsistencies (Gu et al. 2021; Zhao et al. 2023). In audio, neural vocoders introduce spectral discontinuities and unnatural harmonic structures (Sun et al. 2023; Yan et al. 2022). Critically, these artifacts share a common mathematical foundation: they result from imperfect approximation of continuous signals through discrete computational operations.
- *Semantic inconsistency artifacts* stem from the GMs' inability to fully capture complex real-world semantics. Images exhibit physiological impossibilities (asymmetric pupils, inconsistent lighting) (Chen et al. 2021; Amin et al. 2024). In videos, temporal dynamics may be unrealistic, with abrupt motion changes or inconsistent object trajectories (Huda et al. 2024; Sun et al. 2021). Audio displays prosodic anomalies and phonetic inconsistencies (Alzantot et al. 2019; Saleem et al. 2019). These artifacts suggest that detectors focusing on semantic consistency may generalize better across generation techniques than those exploiting low-level statistical patterns.
- *Blending boundary artifacts* occur where synthetic content merges with authentic data. In images and videos, face-swap operations leave unnatural edges or color mismatches around manipulated regions (Frank et al. 2020; Qian et al. 2020; Montserrat et al. 2020; Saikia et al. 2022). Audio exhibits abrupt transitions or spectral discontinuities at edit points (Gao et al. 2021; Wani et al. 2024). These artifacts highlight the challenges of seamless integration between real and synthetic data, emphasizing the need for detectors sensitive to boundary inconsistencies.*Identity consistency as a cross-modal invariant.* While local artifacts (e.g., checkerboard patterns, noise residuals) can be scrubbed by post-processing, identity consistency emerges as a powerful invariant across modalities for DF detection. DF images and videos often fail to preserve the precise biometric geometry of a specific subject across varying poses and lighting conditions (Agarwal et al. 2020; Huang et al. 2023; Cozzolino et al. 2021). Similarly, voice conversion systems often degrade the unique timbre and prosody that define an individual's vocal identity (Alzantot et al. 2019; Saleem et al. 2019). Detectors leveraging identity consistency, through face recognition embeddings in images/videos or speaker verification features in audio, can achieve robust detection that generalizes across generation techniques and resists artifact removal.

Table 5 Summarization of approaches, strengths, and limitations in image modality detection

	Approach	Article	Key idea	Strengths	Limitations
Forensic-based	Physical-based	Chen et al. (2021), Amin et al. (2024), Qiao et al. (2023), Talib et al. (2025)	Color statistics analysis	- Provide strong interpretability for the detection decisions- Don't rely on generator architectures or manipulation techniques	- May become less effective for DM-based images that can preserve semantic content in images Vahdati et al. (2024) - Struggle to low-quality fake images or images applied distortion techniques Barni et al. (2020)
	Physiologically-based	Nirkin et al. (2021), Hu et al. (2021), Wang et al. (2022)	Biological consistency between eyes or facial symmetry		
Data-driven	Conventional classification	Huang et al. (2022), Wang et al. (2022), Mazaheri and Roy-Chowdhury (2022), Katamneni and Rattani (2024), Das et al. (2022), Guo et al. (2023), Tantaru et al. (2024)	Cast as simple binary classification or fine-grained classification problem	- Automatically learn discriminative features from data- End-to-end training	- Rely on the quality and quantity of training data- Limited generalization across different datasets or manipulation techniques- Lack of interpretability since detectors are black-box
	Pixel-level segmentation	Frank et al. (2020), Qian et al. (2020), Dzanic et al. (2020), Liu et al. (2020), Tan et al. (2024), Hong et al. (2024)	Cast as segmentation problem to predict manipulated regions		
	Reconstruction-based learning	He et al. (2021), Cao et al. (2022), Liu et al. (2023), Shi et al. (2023), Guo et al. (2024)	Formulate through the lens of image reconstruction		
Fingerprint-based	Upsampling operation fingerprint	Frank et al. (2020), Qian et al. (2020), Dzanic et al. (2020), Liu et al. (2020), Tan et al. (2024), Tan et al. (2024)	Upsampling operations leave checkerboard patterns in the frequency domain	- Less sensitive to image content since methods focus on model-specific artifacts- Provide clearer reasoning for their decision compared to black-box DL methods	- Generators can remove these fingerprints through additional constraints Chandrasegaran et al. (2021); Neves et al. (2020) - Vulnerable to distortion techniques which can remove these fingerprints

Table 5 (continued)

	Approach	Article	Key idea	Strengths	Limitations
	Denoising process fingerprint	Wang et al. (2023), Ma et al. (2023)	Denoising process leaves lower reconstruction errors between fake and reconstructed images		
Hybrid	Spatial and frequency fusion	Miao et al. (2023), Miao et al. (2022), Wang et al. (2022), Wang et al. (2023)	Leverage spatial and frequency domain	- Capture a broader range of artifacts through different modalities- More robust against single-modality evasion techniques	- Processing multiple feature streams requires more computational resources and may increase inference time- Multi-stream architectures can be harder to train
	Spatial and noise fusion	Kong et al. (2022), Shuai et al. (2023), Guillaro et al. (2023)	Leverage RGB features and noise patterns		

Table 6 Summarization of approaches, strengths, and limitations in video modality detection

	Approach	Article	Key idea	Strengths	Limitations
Forensic-*based*	Physical-based	Xia et al. (2022), Huda et al. (2024)	Color statistics or surface inconsistencies analysis frame-by-frame	- Provide strong interpretability for the detection decisions- Don't rely on generator architectures or manipulation techniques	- May become less effective for DM-based images that can preserve semantic content in images Vahdati et al. (2024)- Struggle with low-quality fake images or images applied distortion techniques Barni et al. (2020)
	Physiologically-based	Sun et al. (2021), Li et al. (2023), Jeon et al. (2022)	Skin color changes, facial movements, and symmetrical face patches		
Data-driven	Frame-based	Wang and Chow (2023), Ciamarra et al. (2024), Ba et al. (2024), Bonettini et al. (2021), Gu et al. (2021), Gu et al. (2022a), Gu et al. (2022b), Gu et al. (2022c)	Based on CNN architectures to classify each frame as real or fake and then fuse them for the final decision	- Take advantage of image-level approaches	- Lack temporal information to detect DFs- Low performance when detecting DF videos Vahdati et al. (2024)
	Video-level	Montserrat et al. (2020), Saikia et al. (2022), Coccomini et al. (2022), Zhang et al. (2022), Zhang et al. (2021), Zheng et al. (2021)	Design architectures or techniques that can capture inter-frame inconsistencies	- Capture both spatial and temporal relationships to identify inconsistencies in video	- High computational and GPU memory requirements for processing multiple frames simultaneously- Identity-driven approaches may struggle when dealing with face reenactment where the identity is preserved throughout the manipulated video
		Agarwal et al. (2020), Huang et al. (2023), Cozzolino et al. (2021), Liu et al. (2023), Dong et al. (2022), Coccomini et al. (2024)	Explore identity inconsistencies over the video		

Table 7 Summarization of approaches, strengths, and limitations in audio modality detection

	Approach	Article	Key idea	Strengths	Limitations
Data-driven	Handcrafted-feature-based	Alzantot et al. (2019), Yang et al. (2018), Xue et al. (2022), Kwak et al. (2021), Gao et al. (2021), Saleem et al. (2019), Wani et al. (2024)	Utilize expert-based audio processing techniques to extract physical and perceptual features for DF audio detection	- Extract predefined features typically requires less computational power- Make detection decisions more transparent and explainable	- Lack of generalization on out-of-domain data Yang et al. (2024)- Predefined features cannot adapt to the specific characteristics of different datasets
	Learnable-feature-based	Jung et al. (2022), Huang et al. (2023), Towards end-to-end synthetic speech detection (2021), Tak et al. (2021), Shin et al. (2024), Chakravarty and Dua (2024), Chen et al. (2024), Yang et al. (2024), Wang and Yamagishi (2024), Tran et al. (2024), Guo et al. (2024), Martín-Doñas et al. (2024), Pan et al. (2024), Doan and Nguyen-Vu (2024), Kim et al. (2024), Cui et al. (2025), Tran et al. (2025), El Kheir et al. (2025), Kim et al. (2025), Tran et al. (2025), Negroni et al. (2025)	Leverage supervised-trainable front-ends or pre-trained self-supervised models to optimal feature representations	- Offer an E2E training process- Capture sophisticated temporal patterns across multiple time scales	- Use SSL models as front-end feature extraction can lead to computational overhead and overfitting to limited downstream data- Learned features often lack clear acoustic meaning, making their decisions harder to explain
Fingerprint-based	Vocoder fingerprint	Yan et al. (2022), Sun et al. (2023)	Explore artifacts left by vocoders	- Provide better explanation for the decisions- Less sensitive to speech content	- Vulnerable to distortion techniques that can remove artifacts

Techniques with limited knowledge transferability. Certain techniques exhibit limited transferability across modalities due to inherent differences in data structure and artifact manifestation:

- *Fingerprint-based methods* are modality-specific, as they target artifacts unique to the generation process of each data type. For instance, upsampling artifacts in images have no direct analogue in audio waveforms, limiting the applicability of such methods across modalities (Frank et al. 2020; Tan et al. 2024; Liu et al. 2020; Dzanic et al. 2020; Qian et al. 2020; Sun et al. 2023; Yan et al. 2022). Meanwhile, vocoder-specific artifacts require audio-specific spectral analysis which are not directly transferable to visual data.
- *Temporal modeling architectures* designed for video DF detection, such as 3D CNNs or LSTMs, exploit frame-to-frame dependencies that have no direct counterpart in static images or 1D audio signals (Montserrat et al. 2020; Saikia et al. 2022, 2022). Conversely, audio-specific architectures like temporal convolutional networks (TCNs) or Recurrent neural networks (RNNs) do not directly apply to spatially structured image data (Jung et al. 2022; Huang et al. 2023).*Cross-modality knowledge transfer.* Some techniques in the visual domain can be successfully adapted to the audio domain. Conversely, certain audio-specific methods hold promise for video DF detection. Specifically:

- *Visual* $\rightarrow$ Audio: Techniques like Vision Transformers (ViT) and Masked Autoencoders (MAE) are now being adapted for audio (e.g., SSAST, Audio-MAE) by treating spectrograms as images (Wani et al. 2024; Saleem et al. 2019). These models leverage self-attention mechanisms to capture long-range dependencies in audio data, similar to their success in image analysis (Cao et al. 2022; Luo et al. 2021; Wang et al. 2022).
- *Audio* $\rightarrow$ Visual: Conversely, techniques native to audio, such as examining phase continuity, are underutilized in video detection. Since video flickering is essentially a "phase discontinuity" in the temporal pixel signal, adapting phase-aware audio architectures to the temporal dimension of video represents a significant, underexplored opportunity for future research.
- *Shared techniques across modalities:* Certain approaches, such as reconstruction-based learning and self-blending augmentation, have demonstrated effectiveness across both visual and audio domains. Reconstruction-based methods that learn to recreate authentic data distributions have been successfully applied to images (He et al. 2021; Cao et al. 2022) and show promise for audio spectrograms (Doan and Nguyen-Vu 2024). Similarly, self-blending augmentation techniques proven effective for images (Cheng et al. 2024) have been successfully transferred to video domains (Wang et al. 2023), indicating potential benefits for audio applications (Martín-Doñas et al. 2024).

However, establishing these cross-modal connections addresses only the architectural design of detectors. While the unimodal and hybrid methods discussed in this section achieve high performance on specific benchmarks, they often operate under the assumption of a closed-set environment. To bridge the gap between these theoretical architectures and practical utility, we must shift our focus to the dynamic conditions inherent in deployment environments.

4 Real-world requirements of effective passive deepfake detection

Building upon the architectural foundations established in Section 3, we now examine the operational criteria necessary for deployment. In addition to high detection accuracy, real-world DF detection systems must meet several critical requirements to be effective in practical applications. These include generalization to unseen data, robustness against adversarial attacks and distortion attacks. In this section, we discuss each requirement in detail, along with current approaches proposed to address them. Figure 6 provides a taxonomy of existing approaches to enhance generalization and robustness of DF detectors. Section 4.1 focuses on generalization, examining five key approaches: data augmentation, designed training strategies, disentanglement learning, unsupervised learning, and adaptive learning. Section 4.2 addresses robustness, covering defenses against adversarial attacks and distortion attacks, as well as techniques for achieving real-world resilience under naturally occurring conditions such as compression, resizing, and audio-visual misalignment.

4.1 Generalization

Generalization refers to the capability of DF detectors to maintain performance when confronted with previously unseen data distributions, manipulation techniques, and generator architectures. This capability is crucial as DF generation techniques continuously evolve. Researchers typically assess generalization through two protocols: (i) Within-domain evaluation and (ii) Cross-domain evaluation. We define five key approaches to enhance generalization, including data augmentation, designed training strategies, disentanglement learning, unsupervised learning, and adaptive learning.

Data augmentation (DA) approaches enhance model generalization by expanding training data diversity. Rather than collecting additional samples, these methods apply various transformations to existing data, creating a broader distribution of examples for the model to learn from. Depending on specific modalities, augmentation operations are various, including (i) image modality: Gaussian blur, JPEG compression, random crop (Ni et al. 2022; Luo et al. 2023; ii) video modality: temporal dropout, temporal repeat, clip-level blending (Wang et al. 2023; iii) audio modality: band-pass filters, time- and pitch-shifting, RawBoost (Doan and Nguyen-Vu 2024; Martín-Doñas et al. 2024). Other works employ more sophisticated DA techniques, including dynamically erasing sensitive facial regions (Wang and Deng 2021; Das et al. 2021) to prevent overfitting and latent space augmentation to simulate variations within forgery features (Yan et al. 2023; Huang et al. 2025). Contrastive learning (CL) has been integrated with DA to eliminate task-irrelevant contextual factors

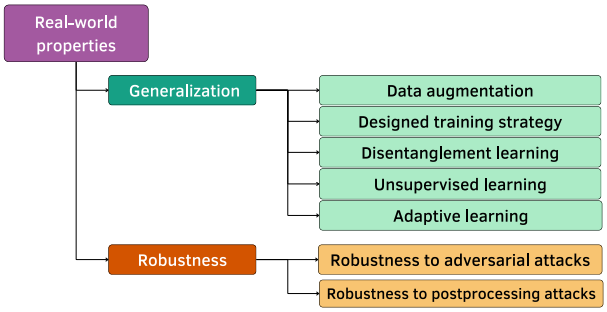


Fig. 6 Taxonomy of generalization and robustness approaches in passive DF detection

(Sun et al. 2022). Kashyap et al. (2025) (Kashyap et al. 2025) utilizes HiFi-GAN for data augmentation and contrastive learning to refine embeddings extracted from self-supervised pre-trained models. This approach enhances the model's ability to distinguish between real and fake audio samples.

Designed training strategy methods structure the learning process to better capture fundamental differences between real and synthetic content. Rather than treating DF detection as a simple binary classification problem, they incorporate inductive biases about the underlying structure of manipulation artifacts into the training process. This helps models learn more abstract, generalizable representations. Choi et al. (2024) (Choi et al. 2024) propose a two-stage training approach that first learns a robust style representation through supervised CL and then integrates these with content features for DF detection. This idea is extended by Zhu et al. (2025) (Zhu et al. 2025) to the audio modality. Cheng et al. (2024) (Cheng et al. 2024) introduce a progressive training strategy that organizes the latent feature space in a progressive manner, establishing a transition from real to fake as following a specific path: Real → Blendfake (self-blended) → Blendfake (cross-blended) → DF. This strategy helps the model learn a more structured representation of forgery artifacts rather than treating each type as disconnected examples. Other works aim to mitigate the dependence on large-scale labeled data by generating synthetic training examples during the training process.Chen et al. (2022) (Chen et al. 2022) leverages the generator-discriminator framework with self-supervised auxiliary tasks to encourage learning generalizable features. This idea is further extended in the audio domain by the work (Febrinanto et al. 2025) which employs a label generattion network to create auxiliary labels to support the training process of the audio DF detector. In contrast, other works produce more challenging forgery artifacts by re-synthesizing real samples (Doan and Nguyen-Vu 2024) or self-blending transformed versions of real samples (Shiohara and Yamasaki 2022; Yan et al. 2024). Hasanaath et al. (2025) (Hasanaath et al. 2025) extends the idea of self-blending by extracting frequency domain features from self-blended images to amplify artifact information. Meanwhile, FreqBlender (Hanzhe et al. 2024) directly generates pseudo-fake faces in the frequency domain by using a decoder to estimate the probability map of the corresponding frequency knowledge.

Disentanglement learning techniques separate content-specific information (e.g., identity, background) from manipulation artifacts. This allows models to focus specifically on learning manipulation-related features without content-specific biases. Liang et al. (2022) (Liang et al. 2022) designs a dual-encoder architecture that separately extracts content and artifact features and employs CL to maximize the separation between these feature spaces. Yan et al. (2023) (Yan et al. 2023) proposes a multi-task disentanglement framework with a contrastive regularization loss for encouraging separation between common and specific forgery features while Guo and Zhen (2023) (Guo and Zhen 2023) constructs a controllable geometric embedding space to decouple irrelevant correlations. DiffusionFake (Chen et al. 2025) forces the detection network to learn disentangled representations of source and target features inherent in DFs by leveraging a pre-trained Stable Diffusion model to reconstruct both source and target identities. This technique helps the model to identify mixed identity information inherent in DFs.

Unsupervised learning methods leverage unlabeled data to learn more generalizable representations. Without relying on explicit labels, which are time-consuming to collect, these approaches focus on learning universal patterns that distinguish authentic from manipulated content. To eliminate the need for pixel-level forgery annotations, UPCL (Zhuang et al.

2022) uses multivariate Gaussian estimation to model the distribution of real and fake image patches in a face region. Unsupervised contrastive learning (UCL) has been employed in (Qiao et al. 2024) to model the feature space based on intrinsic data characteristics, where similar samples are pulled together while dissimilar samples are pushed apart. Several researchers have explored transfer learning from self-supervised representations on generic datasets before fine-tuning for DF detection (Kim et al. 2024; Oorloff et al. 2024).

Adaptive learning approaches enable DF detectors to continuously adapt to new manipulation methods, generator families, or data distributions without compromising previously learned knowledge. Common used techniques include meta-learning (Sun et al. 2021; Han et al. 2023; Zhu et al. 2024; Chen et al. 2022; Xu et al. 2025), incremental learning (Pan et al. 2023), knowledge distillation (Lu et al. 2024; Kim et al. 2025), and domain generalization (Xie et al. 2023). We refer readers to publications (Hospedales et al. 2021; Ven et al. 2022; Gou et al. 2021; Zhou et al. 2022) for a better understanding of these techniques. Instead of fine-tuning the entire model, some studies design an adaptation module to train only a few parameters on new types of fake samples. Examples include leveraging low-rank adaptation matrices (Zhang et al. 2023; Wu et al. 2024) and designing a domain alignment loss on the source and target domains (Seraj et al. 2024). Similarly, Oiso et al. (2024) (Oiso et al. 2024) uses prompt tuning on the target domain to adapt pre-trained models to new target domains by adding learnable prompts to the intermediate feature vectors in the front-end. Yang et al. (2025) (Yang et al. 2025) introduces a RASA framework that projects input features during training and testing phases into a unified style space using a learnable style bank to minimize inter-domain and intra-source discrepancies. This framework minimizes the source risk by incorporating structural empirical risk minimization within a Poincaré ball model to capture the inherent hierarchical relationships of audio data. Yan et al. (2025) (Yan et al. 2025) proposes to use Singular Value Decomposition (SVD) to explicitly decompose a pre-trained model's feature space into two orthogonal subspaces, allowing it to preserve general knowledge while learning new forgery-specific patterns in the residual components. Recently, Nguyen-Le et al. (2025) (Nguyen-Le et al. 2025) introduces an adaptation method that enables backpropagation-based approaches to adapt to new forgery types without requiring any training data. The key novelty is the Uncertainty-aware Negative Learning objective which uses noisy pseudo-labels during online test-time adaptation to encourage the model to explore alternative options rather than reinforcing potentially incorrect initial predictions.

4.2 Robustness

Robustness reflects the detector's ability to maintain reliable performance under various attack scenarios. We identify two critical threats that can compromise DF detectors' resilience: adversarial attacks and distortion attacks. Table 12 summarizes existing methods to improve robustness of DF detectors.

Threat landscape. DF detectors are increasingly vulnerable to adversarial attacks, which introduce carefully crafted perturbations into DF content to deceive detection systems while remaining imperceptible to humans. Recent studies have demonstrated that even SoTA detection models can be significantly compromised by adversarial examples (AEs) (Li et al. 2021; Jia et al. 2022; Neekhara et al. 2021; Hou et al. 2023). In contrast to adversarial attacks, distortion attacks employ various post-processing operations to alter DF quality,

including compression, resizing, noise addition, and color adjustments. (Corvi et al. 2023; Xu et al. 2024) show these techniques effectively obscure generation artifacts, substantially reducing detection performance. These attacks are particularly concerning because they often mirror common media processing operations. This makes them difficult to distinguish from benign transformations. Therefore, it is crucial to develop methods that enhance the detectors' robustness to adversarial attacks and distortion attacks.

Robustness to adversarial attacks. Adversarial training (Bai et al. 2021) is widely adopted to improve model robustness against adversarial attacks by incorporating adversarial examples into the training process. This technique has been demonstrated to be effective across both visual and audio modalities (Devasthale and Sural 2022; Nguyen-Le et al. 2024). Beyond conventional adversarial training, more sophisticated approaches have recently emerged. These include adaptive adversarial training that dynamically adjusts training samples based on attack difficulty (Kawa et al. 2022) and frequency-based adversarial training that uses a generator to produce frequency-level perturbation maps (Jeong et al. 2022). For the black-box setting, D4 (Hooda et al. 2024) presents an ensemble-based method using multiple detector models to exam disjoint parts of the frequency spectrum, forcing attackers to find perturbations effective across multiple frequency ranges. Khan et al. (2024) (Khan et al. 2024) designs an adversarial similarity loss that maintains feature space proximity between original samples and their adversarial versions. These two techniques reduce the space of possible attacks and make successful attacks much harder to generate.

Robustness to distortion attacks. One straightforward approach is to simulate diverse distortion techniques during the training process. Multiple studies have demonstrated that data augmentation significantly improves detection robustness. It exposes models to various distortion techniques they might encounter in real-world scenarios (Luo et al. 2023; Wang et al. 2023; Yan et al. 2023). Beyond simple augmentation, Xu et al. (2024) (Xu et al. 2024) combines a generator with progressive learning strategy to train the models from easy scenarios (non-degraded samples) to increasingly difficult ones (heavily-degraded samples). This method helps the model build robust representations. It also avoids the training instability that can occur when immediately introducing heavily degraded examples. Feng and Owens (2023) (Feng and Owens 2023) cast the problem as anomaly detection where both original DFs and their degraded versions are considered anomalous deviations from authentic content. However, these methods work under the assumption that distortion techniques are well-known, which cannot adopt real-world settings where adversaries can apply unknown techniques. To mitigate this, OST (Chen et al. 2022) applies the test-time-training technique (Liu et al. 2021) that enables detectors to adapt to unknown techniques at inference time.

Real-world resilience represents a critical dimension of DF detection systems that focuses on maintaining reliable performance under naturally occurring real-world conditions. While robustness addresses deliberate adversarial and distortion attacks, real-world resilience concerns the detector's ability to function effectively despite routine transformations media undergoes without malicious intent. ① First, content uploaded to social media platforms (Instagram, TikTok, YouTube) or messaging applications (WhatsApp, Telegram) undergoes automatic compression with varying quality settings based on bandwidth considerations, device specifications, and platform requirements. For instance, YouTube employs different compression algorithms for different resolution options, while WhatsApp significantly reduces image quality during transmission. ② Second, detection systems must handle wide variations in media presentation. Videos uploaded to TikTok may be automatically fitted

with thumbnails that resize content with different aspect ratios, potentially cropping key areas of the frame. Similarly, Instagram Reels limit video length and apply format-specific processing. In audio contexts, platforms like Instagram or Facebook limit clip duration, forcing content to be truncated. ③ Third, videos in the real world often suffer from natural misalignment between their audio and visual streams due to technical issues in encoding and recording processes. A common example is when there is a consistent shift (time delay) of a few frames between what is seen and what is heard throughout the video. This creates a significant challenge for multimodal detectors since they cannot simply flag videos where the audio and visuals do not perfectly sync up. These transformations can inadvertently degrade detection performance by altering or removing subtle manipulation artifacts that detectors rely on.

One straightforward way to address the first problem is to employ multiple models for different quality levels (Liao et al. 2023; Woo 2022; Lee and Woo 2022). However, these methods have significant limitations: (i) Cause computational costs and training data overhead and (ii) Not practical for real-world settings because they require prior knowledge about input video quality to select the appropriate detection model. Le and Woo (2023) (Le and Woo 2023) address these gaps by proposing a universal intra-model collaborative learning framework that enables effective simultaneous detection of different quality DFs using a single model. Differently, Chen et al. (2024) (Chen et al. 2024) uses 3D spatiotemporal trajectories to analyze facial landmark movements across frames with the hypothesis that video compression does not significantly alter the distribution of facial landmarks. In the audio modality, FTDKD (Wang et al. 2024) uses the knowledge distillation technique to help the student model learn lost high-frequency information from the teacher model during compression. Regarding the second problem, the works by Xu et al. (2023, 2024) simulate the thumbnails by using augmentation techniques that mask fixed-size square areas at the same positions within frames. To handle the third problem in multimodal detectors, Feng and Owens (2023) (Feng and Owens 2023) utilizes an autoregressive model to estimate the temporal offset between each video frame and its corresponding audio signal, effectively capturing the time delay distribution.

5 Empirical evaluation and analysis

Existing surveys have extensively reviewed DF detection methods from theoretical and architectural perspectives (Li et al. 2025; Mirsky and Lee 2021; Wang et al. 2022a; Pei et al. 2024; Kaur et al. 2024). However, there remains a critical gap in systematic empirical evaluations of these methods under diverse evaluation protocols. To address this limitation, we conduct comprehensive experiments with three distinct objectives. First, we verify the reported performance of 13 popular unimodal detection methods across image, video, and audio modalities, providing independent validation of their baseline detection capabilities. Second, we systematically evaluate methods (33 methods) specifically designed to enhance generalization, assessing their effectiveness in maintaining performance across different datasets. Third, we examine approaches (6 methods) that claim to improve robustness against adversarial attacks, measuring their resilience when confronted with deliberately crafted adversarial examples. Through this multi-faceted evaluation, we validate claimed improvements in generalization and robustness. We also reveal practical insights into the

trade-offs between detection accuracy, cross-domain performance, and adversarial robustness. These findings provide researchers and practitioners with evidence-based guidance for selecting and developing detection methods that can withstand the evolving challenges of real-world deployment.

The remainder of this section is organized as follows: Section 5.1 describes our selection strategies for choosing representative methods for empirical evaluation. Section 5.2 presents the experimental settings, including datasets and evaluation protocols. Section 5.3 details our implementation configurations. Finally, Section 5.4 presents the evaluation results across three dimensions: reported performance verification (Section 5.4.1), generalization evaluation (Section 5.4.2), and robustness evaluation (Section 5.4.3).

5.1 Selection strategies for empirical evaluation

Given the extensive volume of DF detection methods reviewed in Sections 3 and 5.4.2, it is computationally infeasible to re-implement every method. Therefore, we establish clear selection criteria to identify representative methods for empirical evaluation. Our selection strategy is based on three primary criteria:

1. *Alignment with evaluation objectives:* We select methods that explicitly target the specific evaluation objectives of our study:
    - *Reported performance verification:* To validate the reported accuracy of foundational unimodal detectors, we selected 8 visual methods and 5 audio methods that *focus exclusively on detection tasks*.
    - *Generalization capability evaluation:* To validate claims of cross-domain performance, we selected 33 methods *explicitly designed to improve generalization*. This selection covers diverse categories (e.g., data augmentation, reconstruction learning) to allow for comparative analysis of effectiveness between different methodological categories.
    - *Robustness against adversarial attacks evaluation:* To assess resilience to adversarial examples, we selected 6 methods *specifically proposed to enhance robustness*. This includes both adversarial training techniques and specialized architectures aimed at mitigating adversarial vulnerabilities.
2. *Reproducibility and code availability:* We prioritize methods with publicly available code repositories or released pre-trained models. This ensures the fairness and transparency of our experiments.
3. *Community recognition and impact:* We consider methods that have garnered significant attention in the research community, as indicated by citation counts, adoption in subsequent studies, or inclusion in benchmark evaluations. This criterion helps ensure that our evaluation focuses on influential and widely recognized approaches.

Table 8 summarizes the scope of our empirical evaluation. We evaluate a total of 52 unique methods across three complementary protocols. This ensures comprehensive coverage of both foundational detection capabilities and advanced techniques targeting real-world deployment challenges: generalization and robustness.

Table 8 Summary of the evaluation protocols, number of methods, and their purposes in our empirical study

Evaluation protocol	Number of methods	Purpose
Reported Performance Verification	13 (8 visual + 5 audio)	Validate baseline accuracy claims
Generalization Evaluation	33	Assess cross-domain performance
Robustness Evaluation	6	Test adversarial robustness

All 52 methods selected for empirical evaluation are listed in Table 10 (reported performance verification), Table 11 (generalization evaluation), and Table 12 (robustness evaluation) in Section 5.4. Note that, all these methods were fully re-implemented by the authors using their publicly available code repositories and DeepfakeBench benchmark (Yan et al. 2023) to ensure a fair and standardized comparison across uniform hardware and dataset splits. Other detection methods conceptually reviewed in Sections 3 and 5.4.2 were excluded from the empirical evaluation. These methods were excluded either because they lacked publicly available, reproducible code at the time of this study, or because they fell outside the specific evaluation objectives defined in our selection criteria.

5.2 Experimental settings

5.2.1 Datasets

We utilize 10 widely adopted datasets across image, video, and audio modalities to evaluate the detection methods. To ensure a fair comparison, we select training and testing datasets that are commonly used in the literature. Specifically, for the image modality, we use FF++ (Rossler et al. 2019) as the training dataset, and evaluate methods on DFFD (Dang et al. 2020), Celeb-DF (Li et al. 2020), Wild-DF (Zi et al. 2020), UADFV (Yang et al. 2019) and DFDC (Dolhansky et al. 2019). For the audio modality, we select ASVspoof2019 (Wang et al. 2020) as the training dataset, and utilize ASVSpoof2021 (Liu et al. 2023), WaveFake (Frank and Schönherr 2021), and ITW (Müller et al. 2022) for evaluation. More details about these datasets can be found in Section 2.

5.2.2 Evaluation metrics

We adopt standard metrics to comprehensively evaluate detection performance across different aspects. For assessing the reported performance of detection methods and generalization capability, we employ three widely used metrics: ACC, AUC for visual modality, and EER for audio modality. To quantify robustness against adversarial attacks, we measure the ASR, which indicates the percentage of successful evasions. Regarding video-level methods, metrics are computed at the video level through majority voting aggregation of frame-level predictions, while audio evaluations are performed at the utterance level to maintain consistency with standard benchmarks. More details about these metrics can be found in Section 2.

5.2.3 Evaluation protocols

Detection approaches are evaluated under different distinct protocols to assess their performance, robustness, and generalization capabilities (Yan et al. 2023, 2024). In this work, we

focus on three key protocols: (i) Within-domain evaluation, (ii) Cross-domain evaluation, and (iii) Adversarial attacks evaluation.

① *Within-domain evaluation* examines the accuracy performance of the detector when training and testing on the same dataset. This protocol aims to assess the detection capability under controlled conditions where the manipulation techniques and data distributions are consistent. ② *Cross-domain evaluation* involves evaluating the detector's generalization capability across different/new datasets to handle the distribution shifts in the real-world settings. ③ *Adversarial attacks evaluation* measures the detector's resilience to adversarial examples (AEs). There are three common levels of attacks: (i) white-box attack - attackers have complete knowledge of the model architecture and parameters; (ii) black-box attack - attackers have access only to model outputs; and (iii) transferable attack - the most challenging attack in which attackers can create AEs without needing direct access to the ultimate target model. More details about different types of adversarial attacks in terms of definition and mathematics can be explored in (Li et al. 2024).

5.3 Implementation details

Dataset configuration and roles. Table 9 summarizes the detailed statistics and experimental roles of the ten datasets used in our empirical evaluation. We ensure that training and testing splits are consistent with prior works for fair comparison. Specifically, we use FF++ (Rossler et al. 2019) and ASVspoof 2019 (Wang et al. 2020) as the training datasets for visual and audio modalities, respectively. The remaining datasets serve as unseen target domains to evaluate generalization performance under distribution shifts. No samples from the target domains are seen during the training phase.

Table 9 Detailed statistics and experimental roles of the ten datasets used in empirical evaluation

Dataset	Real samples	Fake samples	No. of train	No. of test	Synthesis methods	Role
FF++ Rossler et al. (2019)	1000	4000	3500	1500	Deepfakes, Face-2Face, Faceswap, NeuralTextures	Training
DFFD Dang et al. (2020)	363	3000	2354	1008	Collected from internet	Testing
Celeb-DF Li et al. (2020)	590	5639	4361	1868	GAN-based models	Testing
Wild-DF Zi et al. (2020)	3805	3509	5119	2195	Collected from internet	Testing
DFDC Dolhansky et al. (2019)	23,654	104,500	89,708	38,446	Collected from internet	Testing
UADFV Yang et al. (2019)	49	49	68	30	FakeApp	Testing
ASVspoof 2019 Wang et al. (2020)	41,913	300,678	239,813	102,778	Neutral TTS, VC (A01-A19)	Training
ASVspoof 2021 Liu et al. (2023)	22,617	589,212	428,281	183,548	TTS, VC, Compression Codecs	Testing
WaveFake Frank and Schönherr (2021)	18,100	117,985	95,259	40,826	MB-MelGAN, FB-MelGAN, HiFi-GAN, WaveGlow, MelGAN, PWG	Testing
ITW Müller et al. (2022)	17,000	14,000	21,700	9300	Collected from internet	Testing

We acknowledge that our experiments primarily utilize conventional benchmark datasets rather than the more challenging datasets discussed in Section 2, such as OpenForensics (Nguyen et al. 2021) for multiple faces, DF-Platter (Narayan et al. 2023) for occlusions, and DF40 (Yan et al. 2024) or DiffusionForensics (Wang et al. 2023) for diffusion-based generation. This decision was guided by three practical considerations. First, we prioritized datasets that are commonly adopted in the existing literature to ensure fair and comparable evaluations with prior methods, as most of these methods report results on the FF++, CelebDF, DFDC, and ASVspoof benchmarks. Second, the majority of publicly available methods with reproducible code were architecturally designed to target artifacts native to GANs and conventional generation pipelines. Evaluating these specific methods on novel diffusion datasets without comprehensive retraining would constitute an extreme out-of-distribution, zero-shot test, thereby unfairly misrepresenting their intended baseline capabilities. Finally, retraining all 52 methods on these emerging challenging datasets would require substantial computational resources beyond the scope of this survey. We believe this gap between the challenging scenarios discussed in Section 2 and the conventional datasets used in our experiments further reinforces our discussion in Section 6, where we highlight these as critical open problems requiring future investigation.

Preprocessing and training settings. Following DeepfakeBench (Yan et al. 2023), we use Dlib library to detect and align faces in visual datasets. Then, we crop aligned faces and resize them to 224×224 resolution, which is the common input size for most visual detection methods.For audio datasets, we extract 16 kHz mono-channel waveforms. Each raw waveform is segmented into 4-second utterances for training and evaluation. In our work, we adopt the same training settings designed by the DeepfakeBench framework (Yan et al. 2023) for fair comparison.

Unless otherwise specified by the original method, we train all models using the Adam optimizer with an initial learning rate of 10^{-4}. We use a batch size of 128 and train for 10 epochs, selecting the best model based on validation performance. All experiments are conducted on NVIDIA 2xH100 GPUs.

Method re-implementation. We re-implement all selected methods based on their publicly available code repositories to ensure a fair comparison.

Adversarial attack settings. For adversarial attacks, we implement white-box, black-box, and transferable attack settings. In the white-box setting, we use the Projected Gradient Descent (PGD) attack (Madry et al. 2017) with an L_∞ norm constraint of $\epsilon = 0.03$, a step size of 0.01, and 40 iterations. For the black-box attack, we follow the work (Hooda et al. 2024) and select two attacks, including the gradient-estimation method (i.e., HSJA (Chen et al. 2020)) and the random search-based method (i.e., SignOPT (Cheng et al. 2019)). To ensure reproducibility, we use the same configuration in Hooda et al. (2024): a maximum query budget of 50,000 queries per image and an L_2 norm perturbation constraint of $\epsilon = 10$. In the transferable attack setting, we employ the Transferable Attack method (Nguyen-Le et al. 2024), generating adversarial examples using a surrogate model and evaluating them on the target model.

5.4 Evaluation results

5.4.1 Reported performance verification

In this experiment, we select 13 popular unimodal methods, including 8 visual methods (UCF (Yan et al. 2023), TALL (Xu et al. 2023), STIL (Gu et al. 2021), SRM (Luo et al. 2021), F3Net (Qian et al. 2020), UIA-ViT (Zhuang et al. 2022), RECCE (Cao et al. 2022), CORE (Ni et al. 2022)) and 5 audio methods (Aasist (Jung et al. 2022), Conformer (Rosello et al. 2023), SCL (Doan and Nguyen-Vu 2024), SSL (Wang and Yamagishi 2024), RawB-Mamba (Chen et al. 2024)). We use 4 datasets for evaluation in each modality, which are used by these methods. For the visual modality, we train all methods on FF++ and evaluate them on Celeb-DF, DFDC, and DFD. For the audio modality, we train all methods on ASVspoof2019 and evaluate them on ASVSpoof21, WaveFake, and ITW. Note that some methods do not report datasets and metrics (denoted by "-") in their original papers, so we only provide our verified results for those cases.

Table 10 shows the comparison results between the reported performance in the original papers and our verified results. We can observe that our verified results are generally lower than the reported performance. Especially, several methods show a significant performance drop, such as UCF (Yan et al. 2023) (AUC drops from 82.4% to 70.61% on Celeb-DF), STIL (Gu et al. 2021) (AUC drops from 89.80% to 66.89% on DFDC), and Conformer (Rosello et al. 2023) (EER increases from 3.28% to 94.82% on ASVSpoof2021). Only a few methods achieve comparable results, such as Aasist (Jung et al. 2022) (EER drops slightly from 0.99% to 0.83% on ASVSpoof19) and RawBMamba (Chen et al. 2024) (EER remains stable at around 1.2% on ASVSpoof19).

The discrepancies can be attributed to several factors: (i) differences in experimental settings, such as data preprocessing, training procedures, and hyperparameter tuning; (ii) variations in dataset splits or versions used for training and testing; (iii) potential overfitting to specific datasets or manipulation techniques in the original studies; and (iv) incomplete reporting of experimental details in some papers, making exact replication challenging. These findings highlight the importance of independent verification of reported results to ensure the reliability and reproducibility of DF detection methods. Moreover, we recommend that future research should utilize DeepfakeBench (Yan et al. 2023) framework to facilitate standardized evaluations and comparisons across studies.

5.4.2 Generalization evaluation

Table 11 summarizes the generalization performance of 33 methods across image, video, and audio modalities. For each method, we report the within-domain (WD) performance on the training dataset and cross-domain (CD) performance on multiple unseen datasets. *Observations.*

Across modalities, most methods reach near-perfect performance in WD settings, but *drop significantly under CD evaluation*. In the visual track, many methods exceed 95% ACC and $98-99\%$ AUC on FF++, yet CD ACC typically falls into the $70-85\%$ band, with only a few approaches sustaining $\approx 88-91\%$ ACC. Among augmentation-based methods, static schemes improve CD but remain moderate (e.g., image/video: $73-83\%$ ACC, AUC $\approx 80-81\%$). Dynamic/semantics-aware augmentation is substantially stron-

Table 10 Verification of reported performance for popular df detection methods. The symbol Δ indicates the performance difference between the reported and our verified results

Method	Dataset	Metric	Reported	Verified	Δ
Visual Methods					
UCF Yan et al. (2023)	FF++	AUC	99.66	98.25	−1.35
		ACC	–	93.51	–
	CelebDF	AUC	82.4	70.61	−11.79
		ACC	–	58.61	–
	DFDC	AUC	80.5	69.95	−10.55
		ACC	–	63.25	–
	DFD	AUC	94.5	77.04	−17.46
		ACC	–	51.31	–
TALL Xu et al. (2023)	FF++	AUC	99.87	96.04	−3.83
		ACC	98.65	91.80	−6.85
	CelebDF	AUC	90.79	84.04	−6.75
		ACC	–	77.58	–
	DFDC	AUC	76.78	74.05	−2.73
		ACC	–	65.19	–
	DFD	AUC	–	85.55	–
		ACC	–	83.46	–
STIL Gu et al. (2021)	FF++	AUC	99.64	96.16	−3.48
		ACC	–	90.57	–
	CelebDF	AUC	75.58	70.49	−5.09
		ACC	–	62.46	–
	DFDC	AUC	89.80	66.89	−22.91
		ACC	–	60.09	–
	DFD	AUC	–	77.11	–
		ACC	–	71.20	–
SRM Luo et al. (2021)	FF++	AUC	99.4	98.23	−1.17
		ACC	97.1	95.15	−1.95
	CelebDF	AUC	79.4	56.37	−23.03
		ACC	–	55.52	–
	DFDC	AUC	79.7	77.06	−2.64
		ACC	–	71.22	–
	DFD	AUC	91.9	85.88	−6.02
		ACC	–	85.83	–
F3Net Qian et al. (2020)	FF++	AUC	99.9	98.58	−1.32
		ACC	99.9	95.36	−4.54
	CelebDF	AUC	–	58.64	–
		ACC	–	56.06	–
	DFDC	AUC	–	71.25	–
		ACC	–	64.67	–
	DFD	AUC	–	84.58	–
		ACC	–	77.28	–
UIA–ViT Zhuang et al. (2022)	FF++	AUC	99.33	94.06	−5.27
		ACC	99.15	89.15	–10
	CelebDF	AUC	84.5	69.25	−15.25
		ACC	–	64.09	–
	DFDC	AUC	75.80	72.43	−3.37
		ACC	–	60.97	–

Table 10 (continued)

Method	Dataset	Metric	Reported	Verified	Δ
	DFD	AUC	94.68	76.23	−18.45
		ACC	–	70.51	–
RECCE Cao et al. (2022)	FF++	AUC	0.9932	0.9805	−1.27
		ACC	0.9706	0.9351	−3.55
	CelebDF	AUC	0.6871	0.5047	−18.24
		ACC	–	0.5705	–
	DFDC	AUC	0.6906	0.7162	+2.56
		ACC	–	0.6545	–
	DFD	AUC	–	0.8122	–
		ACC	–	0.8519	–
CORE Ni et al. (2022)	FF++	AUC	1.0	0.9793	−2.07
		ACC	0.9997	0.9249	−7.48
	CelebDF	AUC	0.7571	0.6610	−9.61
		ACC	–	0.6349	–
	DFDC	AUC	0.7241	0.7125	−1.16
		ACC	–	0.6467	–
	DFD	AUC	0.94	0.8588	−8.12
		ACC	–	0.8718	–
Audio Methods					
Aasist Jung et al. (2022)	ASVSpoof-19-LA	EER	0.99	0.83	−0.16
	ASVSpoof-21-DF	EER	–	23.64	–
	WaveFake	EER	–	39.01	–
	ITW	EER	–	56.45	–
Conformer Rosello et al. (2023)	ASVSpoof-19-LA	EER	0.97	0.95	−0.02
	ASVSpoof-21-DF	EER	3.28	94.82	+91.54
	WaveFake	EER	–	68.17	–
	ITW	EER	–	56.45	–
SCL Doan and Nguyen-Vu (2024)	ASVSpoof-19-LA	EER	2.88	3.33	+0.45
	ASVSpoof-21-DF	EER	2.17	2.87	+0.7
	WaveFake	EER	–	22.91	–
	ITW	EER	4.51	4.55	0.04
SSL Wang and Yamagishi (2024)	ASVSpoof-19-LA	EER	1.91	2.66	+0.75
	ASVSpoof-21-DF	EER	5.67	5.72	+0.05
	WaveFake	EER	1.3	1.99	+0.69
	ITW	EER	6.1	5.26	−0.84
RawBMamba Chen et al. (2024)	ASVSpoof-19-LA	EER	1.19	1.2	+0.01
	ASVSpoof-21-DF	EER	15.85	49.99	+34.14
	WaveFake	EER	–	35.45	–
	ITW	EER	–	48.16	–

ger. For example, representative-based erasing (Wang and Deng 2021) retains 80.78% ACC and 82.03% AUC, and frequency-domain self-blending (FreqBlender) (Hanzhe et al. 2024) attains 83.01% ACC and 85.18% AUC. Designed training strategies yield mixed results, with frequency-domain self-blending (Hanzhe et al. 2024) emerging as the most effective technique (89.01% ACC, 89.18% AUC in CD). This superior performance suggests that manipulating features in the frequency domain helps models learn more universal artifacts.

Table 11 Evaluation of deepfake detection generalization methods on within-domain and cross-domain protocols

Approach	Article	Key Idea	Training Dataset	Evaluation							
				WD				CD			
				Dataset	ACC (%)	AUC (%)	EER (%)	Dataset	ACC (%)	AUC (%)	EER (%)
Data Augmentation	Luo et al. (2023)	Static image augmentation	②	②	94.57	97.87	–	③, ④, ⑤, ⑥, ⑦	73.21	81.43	–
	Doan and Nguyen-Vu (2024)	Static audio augmentation	①	①	–	–	0.15	⑧, ⑨, ⑩	–	–	4.76
	Martín-Doñas et al. (2024)	Static audio augmentation	①	①	–	–	0.22	⑧, ⑨, ⑩	–	–	11.49
	Wang et al. (2023)	Static video augmentation	②	②	98.03	99.4	–	③, ④, ⑤, ⑥, ⑦	78.14	80.07	–
	Das et al. (2021)	Dynamic image augmentation	②	②	98.24	96.73	–	③, ④, ⑤, ⑥, ⑦	69.59	71.01	–
	Wang and Deng (2021)	Dynamic image augmentation	②	②	99.34	99.97	–	③, ④, ⑤, ⑥, ⑦	80.78	82.03	–
	Yan et al. (2023)	Image-latent space augmentation	②	②	97.54	92.34	–	③, ④, ⑤, ⑥, ⑦	83.45	81.06	–
	Huang et al. (2025)	Audio-latent space augmentation	①	①	–	–	1.9	⑧, ⑨, ⑩	–	–	5.76
	Sun et al. (2022)	Dual-contrastive learning	②	②	98.71	99.3	–	③, ④, ⑤, ⑥, ⑦	82.16	80.45	5.76
Designed Training Strategy	Chen et al. (2022)	GAN framework + self-supervised auxiliary tasks	②	②	97.13	98.21	–	③, ④, ⑤, ⑥, ⑦	84.86	76.89	–
	Shiohara and Yamasaki (2022)	Self-blending technique	②	②	93.21	99.65	–	③, ④, ⑤, ⑥, ⑦	84.35	81.35	–
	Hasanaath et al. (2025)	Self-blending frequency technique	②	②	83.15	86.31	–	③, ④, ⑤, ⑥, ⑦	73.15	70.11	–
	Hanzhe et al. (2024)	Self-blending in frequency domain	②	②	97.56	96.13	–	③, ④, ⑤, ⑥, ⑦	83.01	85.18	–
	Yan et al. (2024)	Video-level self-blending technique	②	②	97.8	95.23	–	③, ④, ⑤, ⑥, ⑦	81.03	83.21	–
	Zhu et al. (2025)	Two-stage training to learn style and linguistic representations	①	①	–	–	2.0	⑧, ⑨, ⑩	–	–	9.2
	Choi et al. (2024)	Two-stage training to learn style representation	②	②	94.21	99.1	–	③, ④, ⑤, ⑥, ⑦	82.9	84.78	–
Disentanglement Learning	Liang et al. (2022)	Dual-encoder architecture	②	②	97.83	94.5	–	③, ④, ⑤, ⑥, ⑦	72.3	76.83	–

Table 11 (continued)

Approach	Article	Key Idea	Training Dataset	Evaluation							
				WD				CD			
				Dataset	ACC (%)	AUC (%)	EER (%)	Dataset	ACC (%)	AUC (%)	EER (%)
	Guo and Zhen (2023)	Model a controllable geometric embedding space	②	②	93.43	98.4	–	③, ④, ⑤, ⑥, ⑦	73.14	83.31	–
	Chen et al. (2025)	Identity disentanglement by reversing the generative process	②	②	94.23	97.5	–	③, ④, ⑤, ⑥, ⑦	80.5	83.78	–
Unsupervised Learning	Zhuang et al. (2022)	Multivariate Gaussian estimation	②	②	92.15	99.33	–	③, ④, ⑤, ⑥, ⑦	79.45	84.87	–
	Qiao et al. (2024)	Unsupervised contrastive learning	②	②	97.45	99.24	–	③, ④, ⑤, ⑥, ⑦	81.98	82.56	–
	Kim et al. (2024)	Transfer learning from self-supervised representations	①	①	–	–	1.02	⑧, ⑨, ⑩	–	–	7.31
Adaptive Learning	Han et al. (2023)	Meta learning	②	②	97.0	98.1	–	③, ④, ⑤, ⑥, ⑦	78.26	82.01	–
	Zhu et al. (2024)	Meta learning	②	②	95.6	97.24	–	③, ④, ⑤, ⑥, ⑦	68.67	75.97	–
	Chen et al. (2022)	Meta-learning + test-time training	②	②	92.56	98.04	–	③, ④, ⑤, ⑥, ⑦	69.7	82.1	–
	Pan et al. (2023)	Incremental learning + knowledge distillation	②	②	98.55	95.53	–	③, ④, ⑤, ⑥, ⑦	70.78	78.9	–
	Xie et al. (2023)	Source domains aggregation + triplet loss	①	①	–	–	4.21	⑧, ⑨, ⑩	–	–	17.06
	Lu et al. (2024)	One-class knowledge distillation	①	①	–	–	1.78	⑧, ⑨, ⑩	–	–	8.36
	Zhang et al. (2023)	Low-rank adaptation matrices	①	①	–	–	6.51	⑧, ⑨, ⑩	–	–	3.28
	Wu et al. (2024)	Low-rank adaptation matrices	①	①	–	–	1.98	⑧, ⑨, ⑩	–	–	11.25
	Oiso et al. (2024)	Prompt tuning	①	①	–	–	6.43	⑧, ⑨, ⑩	–	–	9.05
	Yan et al. (2025)	Orthogonal weight modification	②	②	97.61	99.0	–	③, ④, ⑤, ⑥, ⑦	89.15	91.72	–
	Nguyen-Le et al. (2025)	Test-time adaptation and negative learning	②	②	94.26	96.14	–	③, ④, ⑤, ⑥, ⑦	89.05	90.25	–

Notes: • i) *Training Dataset*: The dataset that the method trained on • ii) *Evaluation*: Protocols that the method evaluated on: Within-Domain (WD), Cross-Domain (CD) • iii) *Numbers* denotes the datasets used for training and evaluation: ASVspoof19 (①), FF++ (②), CelebDF (③), DFDC (④), DFFD (⑤), WildDF (⑥), UADFV (⑦), ASVSpoof21 (⑧), WaveFake (⑨), ITW (⑩)

Disentanglement methods (Liang et al. 2022; Guo and Zhen 2023; Chen et al. 2025) generally underperform compared to other categories, with the best method (Chen et al. 2025) achieving only 80.5% ACC in CD evaluation. This indicates that while disentangling content and artifact features is conceptually appealing, practical implementations often struggle with imperfect separation, leading to overfitting to content biases. Unsupervised learning methods show promise but with high variance in performance. Transfer learning from self-supervised representations (Kim et al. 2024) achieves competitive results in audio (7.31% EER), while unsupervised contrastive learning (Qiao et al. 2024) maintains reasonable visual detection performance (81.98% ACC). Adaptive learning methods show the most significant improvements in CD performance. Recent methods, including orthogonal weight modification (Yan et al. 2025) and test-time adaptation (Nguyen-Le et al. 2025), deliver substantial boosts without requiring full retraining, achieving 89.15% and 89.05% ACC respectively in CD settings.

Insights. From these results, several key insights emerge:

- Methods leveraging frequency domain information consistently outperform spatial-only approaches. This indicates that artifacts left by GMs are more stable across different datasets.
- While data augmentation provides modest improvements at low computational cost, adaptive learning methods achieve superior generalization at the expense of increased complexity and runtime adaptation requirements. This trade-off must be carefully considered for real-world deployment scenarios.
- AUC-ACC mismatch indicates calibration drift. Several methods show high AUC with lower ACC in CD, implying threshold miscalibration under shift. Post-hoc calibration or domain-aware thresholding is recommended for deployment.
- Audio detection methods generally exhibit larger performance drops in CD settings compared to visual methods. This suggests that audio artifacts may be more sensitive to dataset-specific characteristics, necessitating further research into robust audio feature representations.

These findings underscore a critical challenge for the field: despite significant research efforts, the generalization gap remains substantial enough to limit practical deployment of DF detectors. The consistent $15-20\%$ performance drop across domains indicates that current detection methods remain vulnerable to novel generation techniques, highlighting the urgent need for fundamentally new approaches that can learn truly invariant forgery representations.

5.4.3 Robustness evaluation

We evaluate six methods against three common adversarial settings, including white-box (WA), black-box (BA), and transferable attacks (TA), using Attack Success Rate (ASR; lower is better). Table 12 shows all defenses reduce ASR, but the magnitude and consistency vary notably across threat models.

Attack Severity Analysis. The results reveal a clear vulnerability level of detection methods to adversarial attacks. WA is consistently the most effective, achieving ASR of $81.06-95.62\%$ before defense, indicating that full model knowledge successfully evade

Table 12 Evaluation of deepfake detection robustness methods against adversarial attacks

Approach	Article	Key Idea	Attack Type	Evaluation (ASR (%))	
				Before defense	After defense
Robustness to adversarial attacks	Devasthale and Sural (2022)	Purely adversarial training	WA	81.06	37.03
	Nguyen-Le et al. (2024)	Purely adversarial training	TA	34.17	2.01
	Jeong et al. (2022)	Frequency Perturbation	WA	87.69	40.67
	Kawa et al. (2022)	Adaptive adversarial training	WA	85.49	5.25
	Hooda et al. (2024)	Disjoint frequency ensemble	BA	74.05	27.54
	Khan et al. (2024)	Adversarial similarity loss	WA	95.62	25.38

detection in approximately 9 out of 10 cases. This severity stems from the attacker's ability to compute gradients and craft optimal perturbations. These perturbations are tailored to the target model's decision boundaries. BAs demonstrate moderate threat levels with a 74.05% ASR. Transferable attacks show the lowest baseline ASR at 34.17%, suggesting that generating effective AEs without direct access to the target model is more challenging. This is because gradient obfuscation and model mismatch significantly limit attack success.

Defense Effectiveness Analysis. Table 12 shows that detection methods exhibit varying degrees of robustness improvements across different defense strategies. Pure adversarial training (Devasthale and Sural 2022; Nguyen-Le et al. 2024) cuts WA ASR from 81.06% to 37.03% and TA ASR from 34.17% to 2.01%, demonstrating that exposure to AEs during training enhances model robustness. However, the post-defense WA ASR remains relatively high at 37.03%. This indicates that static adversarial training alone is insufficient for robust defense against fully informed attackers. Adaptive adversarial training (Kawa et al. 2022) demonstrates exceptional performance against WAs, reducing ASR from 85.49% to 5.25% . This dramatic reduction suggests that dynamically adjusting training samples based on attack difficulty enables models to learn more robust decision boundaries. For BA, disjoint frequency ensembles (Hooda et al. 2024) lower ASR from 74.05% to 27.54%, indicating that leveraging diverse spectral features can improve robustness in query-limited scenarios.

Insights. Several key insights emerge from this evaluation:

- No single defense method provides universal protection across all attack types. Therefore, it is important to adopt a multi-faceted approach, combining adversarial training, input transformations, and ensemble strategies to enhance overall robustness.
- The substantial gap between WA ($> 80\%$ ASR) and TA success rates (34.17%) suggests that many detectors inadvertently rely on gradient masking rather than learning genuinely robust features. This vulnerability becomes critical when attackers gain model access.
- While adaptive method (Kawa et al. 2022) achieve superior robustness, they require sophisticated training procedures and increased computational overhead. This trade-off between robustness and efficiency must be balanced based on deployment constraints.

5.4.4 Empirical limitations

The empirical evaluation in this survey is intended as a standardized descriptive comparison rather than an inferential statistical benchmark. We therefore report the observed ACC, AUC, EER, and ASR values under fixed training/testing splits and unified implementation settings, but we do not claim statistically significant superiority among individual methods. We acknowledge that confidence intervals, multi-seed variance analysis, and formal significance testing would provide additional statistical evidence. However, applying such analysis comprehensively across 52 methods, 10 datasets, multiple modalities, and several evaluation protocols would require substantial additional statistical analysis and repeated runs beyond the scope of this survey. Accordingly, the empirical findings should be interpreted as observed trends under a unified protocol, while full statistical reliability analysis remains an important direction for future benchmark-oriented studies.

6 Challenges and future directions

Despite significant advances in DF detection research, several critical challenges remain unresolved. These challenges hinder the practical deployment of detection systems. This section discusses six key areas that require further study. We begin with the challenge of detecting DFs under challenging real-world conditions such as occlusion and multiple faces (Section 6.1). followed by the need for generalization to emerging generation techniques (Section 6.2). We then examine the trade-off between generalization and robustness (Section 6.3), and dataset imbalances and biases (Section 6.4). Finally, Finally, we discuss computational efficiency requirements for edge deployment (Section 6.5), and privacy-preserving detection mechanisms (Section 6.6).

6.1 Detection of DFs in challenging conditions

Current approaches often focus on fully synthetic content and demonstrate limited effectiveness in identifying DFs under challenging conditions, including face occlusion, multiple facial DFs, and partial manipulations.

Face occlusion. Advanced techniques, such as 3D face restoration (Chen et al. 2024) and blind inpainting (Criminisi et al. 2004), have successfully addressed occlusion challenges in generic face recognition. There exists significant potential to adapt and integrate these techniques into DF detection frameworks to maintain performance when faces are partially obscured or occluded.

Multiple facial DFs. Researchers evaluate their detection methods on specialized datasets designed for multi-subject forgery detection, including OpenForensics (Nguyen et al. 2021) and DF-Platter (Narayan et al. 2023). These datasets specifically address the increasingly common scenario where multiple subjects within a single frame/image have been manipulated. This presents distinct challenges beyond single-face DF detection.

Partial manipulation. Detection systems must evolve to identify localized manipulations that alter only specific facial regions, partial frames, or partial utterances. Developing patch-based architectures VisionTransformer like (Khan et al. 2022) and implementing

region-specific attention mechanisms could improve the detection of these increasingly sophisticated partial manipulations.

6.2 Generalization to emerging generation techniques

Our empirical evaluation in Section 5.4.2 reveals that existing DF detectors exhibit significant performance degradation when confronted with unseen generation methods across conventional datasets. Because our baseline experiments demonstrate a substantial generalization gap (10–15% degradation) even within GAN-dominated distributions, we project this vulnerability will be exponentially more severe when applied to emerging Diffusion Models (DMs). With the rapid evolution of GMs, detectors must adapt to novel architectures and training paradigms that may introduce entirely new artifact patterns. This necessitates the development of detection approaches that can learn invariant features. These features must be robust to a wide spectrum of generation techniques. *(1) Foundation model approaches* that leverage large-scale pre-training on diverse synthetic data to learn generalizable representations. *(2) Continual learning frameworks* that enable detectors to incrementally update their knowledge base as new generation methods emerge, without catastrophic forgetting of previously learned artifacts. *(3) Generation-aware meta-learning strategies* that explicitly models the space of possible generation techniques rather than memorizing specific artifacts.

6.3 Generalization and robustness trade-off.

Research from the broader computer vision field suggests that there exists an inherent trade-off between a model's ability to generalize to unseen data and its robustness against adversarial perturbations (Zhang et al. 2019; Stutz et al. 2019; Li and Xu 2023; Gowda et al. 2024). In the context of DF detection, this trade-off may also exist. Models optimized for high generalization may inadvertently become more susceptible to adversarial attacks, and vice versa. Understanding and managing this trade-off is crucial for developing practical DF detectors. These detectors must operate effectively in real-world scenarios where both unseen generation techniques and adversarial threats are prevalent. Future research should focus on (1) *Theoretical analyses* to formally characterize the relationship between generalization and robustness in DF detection, identifying conditions under which improvements in one aspect may lead to degradation in the other. (2) *Multi-objective optimization* that explicitly balances generalization and robustness during training, allowing deployment-specific trade-off selection.

6.4 Dataset imbalances and biases

DF detectors trained on current datasets face significant performance degradation when deployed in real-world settings due to class imbalance issues (Layton et al. 2024). These detectors exhibit a bias toward classifying content as fake when confronted with realistic deployment scenarios. In such scenarios, authentic content vastly outnumbers DFs. This imbalance leads to excessive false positive rates that undermine the practical utility of detection systems. Long-tailed recognition is a potential direction to improve the accuracy of the "real" class with the least influence on the "fake" class (Miao et al. 2024; Bai et al.

2023). Additionally, researchers should consider reformulating DF detection as an anomaly detection problem. In this formulation, models learn a representation of authentic content variation rather than decision boundaries between real and fake classes (Ho et al. 2024; Salehi et al. 2022). This approach naturally accommodates imbalanced scenarios by focusing on characterizing normal content patterns.

6.5 Computation efficiency for edge deployment

Many advanced detection approaches incur significant computational overhead, making them impractical for real-time or resource-constrained applications. For example, large-scale SSL models used as feature extractors demand substantial memory and processing power. This hinders deployment on edge devices or mobile platforms (Wu et al. 2024). Future research should prioritize the development of lightweight architectures and efficient training paradigms that maintain high detection performance while minimizing computational requirements. Techniques such as model pruning (He and Xiao 2023), quantization (Egashira et al. 2024), and knowledge distillation (Gou et al. 2021) can be leveraged to create compact yet effective DF detectors suitable for deployment in diverse environments.

6.6 Privacy-preserving detection

Current detection approaches often require access to full media content or datasets for training, raising significant privacy concerns when deployed at scale across communication platforms or personal content. Federated learning (Zhang et al. 2021) is a distributed machine learning paradigm. It enables multiple models to collaboratively train a global model without sharing their raw data. This framework should be adapted for DF detection to enable detector training and updating without centralizing sensitive media. Furthermore, researchers should consider developing privacy-preserving feature extraction techniques that convert media into non-reversible representations before analysis. These approaches would enable detection systems to operate on transformed data. Such data preserves manipulation artifacts while removing personally identifiable information (e.g., speech content or identity information).

7 Conclusion

This survey has systematically analyzed the inter-modality relationships between unimodal DF detection methods across image, video, and audio domains. We also reviewed state-of-the-art techniques for enhancing detection generalization and robustness against adversarial attacks. Our empirical evaluations of 50 methods across 10 datasets reveal critical insights. First, there exists a gap between reported performance and our validated results. This underscores the need for future research to adopt standardized evaluation protocols. Second, while many methods achieve near-perfect accuracy in within-domain settings, current detectors exhibit a persistent $15 - 20\%$ performance degradation in cross-domain scenarios. This indicates fundamental limitations in learning invariant forgery representations. Third, adversarial attacks pose a significant threat to detection reliability. White-box attacks achieve over 80% success rates against undefended models. Although adaptive adversarial

training can substantially mitigate this vulnerability, it introduces considerable computational overhead. Our analysis also reveals critical dataset limitations, including severe class imbalance and the absence of adversarial robustness benchmarks. These limitations constrain the development of practical detection systems.

Despite significant research efforts, DF detection remains an open problem. The rapid evolution of GMs demands a paradigm shift from pattern memorization toward learning truly invariant forgery representations. Achieving reliable real-world deployment requires fundamental advances in both theoretical understanding and practical implementation.

Author contributions H-HN-L conducted a systematic review, analyzed trends and relationships, and wrote the main manuscript V-TT conducted empirical experiments, and drew all the figures D-TN and N-AL-K supervised the project, provided critical feedback, and revised the manuscript

Funding Open Access funding provided by the IReL Consortium. This publication has emanated from research conducted with the financial support of Science Foundation Ireland under Grant number 18/CRT/6183.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

Informed consent Not applicable.

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